EE 571 PROJECT REPORT

ANALYSIS OF THE DIRICHLET PROCESS AS A BAYESIAN NON-PARAMETERIC PRIOR IN MIXTURE OF MODELS PERSPECTIVE

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1 Introduction

Discovering latent structure in data is a fundamental problem in signal processing, whose solutions reside under the realm of statistics and whose applications extend to machine learning, computer vision, communications, and biomedical signal analysis. In many applications, such as clustering and density estimation, observations are assumed to arise from a mixture of underlying components whose identities are unobserved, leading to the concept of latent variables. Mixture models provide a principled statistical framework for such problems by representing data as samples generated from a weighted combination of distributions in question.

Classical mixture models, such as well-known Gaussian Mixture Models (GMMs), require the number of mixture components to be specified and fixed in advance. In practice, this assumption may fall short, since the true number of latent clusters is unknown and must be in alignment with what the data demands. Model selection techniques can be employed to address this limitation, but they usually introduce some metric (e.g. Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC)) to select the number of mixture components, requiring multiple fits, increasing complexity.

Bayesian nonparametric methods provide an alternative approach by allowing the model complexity to grow with the data. Among these methods, the Dirichlet Process (DP) plays a role as a prior over probability measures. Rather than fixing the number of mixture components, the Dirichlet Process defines a distribution over distributions, enabling the number of clusters to be inferred implicitly through Bayesian inference.

The objective of this project is to present a detailed analysis of the Dirichlet Process by building its theory along with its applied perspective. Starting from its measure-theoretic foundations, the mathematically formal definition of the Dirichlet Process is introduced, complying with Ferguson's formulation. The constructive stick-breaking representation is then discussed, leading naturally to the Chinese Restaurant Process as a predictive mechanism for data partitioning problem. Furthermore, the connection between the Dirichlet Process and mixture models is stated, illustrating how Dirichlet Process Mixture Models emerges as the infinite-limit counterpart of finite mixture models. Finally, experimental results on synthetic data are presented to demonstrate the clustering capability of the mixture models built upon DP, along with the comparison with the parameter estimation based finite mixture models (e.g. Gaussian Mixture Model (GMM)).

2 Fundamentals of Measure Theory

The Dirichlet Process is defined with the notion of distribution over probability measures. Hence, its formulation necessarily relies on a rigorous mathematical definition of what a probability measure is and on which objects such measures are allowed to act. Without this mathematical framework, the notion of placing a probability distribution on an infinite-dimensional space of distributions would be ill-defined. Measure theory provides the language and structure needed to define probability measures on abstract spaces and to reason about random measures in a consistent way. For this reason, we begin by reviewing the measure-theoretic foundations of probability.

A measure is a rule that assigns a non-negative so-called 'size' to sets. Its space consists of the following components $(X, \mathcal{A}, \mu)$ with

- X : the space (objects)

- $\mathcal{A}$: a σ-algebra on X (defining subsets are allowed for measure)

- μ : a function that assign sizes

$$\mu : \mathcal{A} \to [0, \infty]$$

A measure is defined on a σ-algebra, which specifies the collection of measurable subsets of X

2.1 σ-Algebra

Let X be a set. A collection $\mathcal{A} \subseteq \mathcal{P}(X)$ (where $\mathcal{P}$ denotes the power set) is a σ-algebra if it satisfies the following (as also described in Billingsley, 1995, Section 1.2) [1]:

1) The whole space is included: $X \in \mathcal{A}$

2) It is closed under the complements: $A \in \mathcal{A} \Rightarrow A^c \in \mathcal{A}$

3) Closed under the countable unions: $A_1, A_2, \cdots \in \mathcal{A} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$

2.1.1 Borel σ-Algebra

The Borel σ-algebra on $\mathbb{R}$ is defined as the smallest σ-algebra containing all open sets. (Not every set on real numbers can construct a σ algebra such as Vitali Sets [2]). Equivalently, it is the σ-algebra generated by open intervals (a, b). Concretely, Borel sets are obtained by starting from all open intervals and closing under complements and countable unions (and hence countable intersections). These "moves" are to satisfy the properties of σ-algebra. Hence, it creates measurable sets using the real numbers, which will (in the end) allow us to define the **random variables** as measurable maps

2.2 Properties of Measures and Probability Measure

A measure with the mapping $\mu : \mathcal{A} \to [0, \infty]$ should satisfy the following properties:

1) Non-negativity: $\mu(A) \geq 0$

2) Empty set must be assigned to mass of 0: $\mu(\emptyset) = 0$

3) Countable additivity: If $A_1, A_2, \ldots$ are disjoint, $\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu\left(A_n\right)$

A useful example of measures is the **Lebesgue measure** with the following properties:

- $\lambda([a, b]) = b - a$

- Translation invariant:$\lambda(A + x) = \lambda(A)$

- Scaling: $\lambda(cA) = |c|\lambda(A)$

- $\lambda(\mathbb{R}) = \infty$

We will use the Lebesgue measure in the probability discussion. In fact, Lebesgue measurability defines a Lebesgue σ-algebra of which the Borel σ-algebra is a subset $\mathcal{B}(\mathbb{R}) \subset \mathcal{L}$ [3] and Lebesgue σ-algebra is one of the completion of the Borel σ-algebra .

2.2.1 Probability Measure

In its most simplest terms, it is a measure that sums up to 1: $P(\Omega) = 1$ and onto that, the so-called probability space is $(\Omega, \mathcal{F}, P)$ where Ω is the sample space $\mathcal{F}$ is σ-algebra of events (we must restrict to the σ-algebra and P is the probability measure.

$$P : \mathcal{F} \to [0, 1]$$

Hence for $A \in \mathcal{F}$, $P(A)$ is "how likely the event A is".

Random variable is in fact a function from the sample space to real numbers $X : \Omega \to \mathbb{R}$. For us to ensure the measure property of probability notion $P(X \in B)$, we need " $X \in B$ to be an event in $\mathcal{F}$. With the definition:

$$\{\omega \in \Omega : X(\omega) \in B\} = X^{-1}(B)$$

we require for measurability property

$$X^{-1}(B) \in \mathcal{F} \quad \text{for all Borel sets } B \subseteq \mathbb{R}.$$

by doing so our probability measure can be turned into the one denoted as

$$\mu_X : \mathcal{B}(\mathbb{R}) \to [0, 1]$$

satisfying $\mu_X(B) := \mathbb{P}\left(X^{-1}(B)\right), \quad B \in \mathcal{B}(\mathbb{R})$

Hence a random variable transfers the probability measure from one space to another.

Relation Between Probability Measure and the Lebesgue Measure

Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. The governing law (distribution) of X is the measure:

$$\mu_X(A) = \mathbb{P}(X \in A), \quad A \in \mathcal{B}(\mathbb{R}).$$

If $\mu_X \ll \lambda$, is satisfied (The measure μ is absolutely continuous with respect to λ if for each A in $\mathcal{F}, \lambda(A) = 0$ implies $\mu(A) = 0$. In this case μ is also said to be dominated by λ, and the relation is indicated by $\mu \ll \lambda$ [1]) Radon–Nikodym theorem claims that there exists an f satisfying

$$\mu_X(A) = \int_A f(x) d\lambda(x)$$

which is indeed called the density function on the Lebesgue measure and in measure theory notation $f = \frac{d\mu_X}{d\lambda}$. Common probability densities and integrations are based on the existence of such a density function defined with the condition above. Hence, we can interpret this concept as a mind-map as:

$$\text{Lebesgue measure} \xrightarrow{\text{multiply by density}} \text{Probability measure}$$

However, not all probability measures yield density since Lebesgue measure forces 0 mass on singletons on the real numbers but a probability measure can put a non zero mass on singletons such as point mass Dirac case:

$$\delta_{x_0}(A) = \begin{cases} 1, & x_0 \in A \\ 0, & x_0 \notin A. \end{cases}$$

In that case, we cannot find densities (e.g. we can still do integration/expectations as $\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) d\mu_X(x)$ but we cannot write in the common $\int g(x) f(x) dx$ integration because there is no such f.)

2.2.2 Generalization of the Probability Measure

Now, we can generalize the probability definition and the random variable concept and measure mapping $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ to a more general parameter/state space Θ. (we will lay out the details how to pick a σ-algebra $\mathcal{G}$ on Θ ($\mathcal{B}(\Theta)$ is often available). We define a "random element" $\theta : (\Omega, \mathcal{F}) \to (\Theta, \mathcal{G})$ and the measurability implies $\theta^{-1}(A) \in \mathcal{F}$ for all $A \in \mathcal{G}$. Hence, the mind-map leads us to define the probability measure on that space:

$$\mu_\theta(A) = \mathbb{P}(\theta \in A) = \mathbb{P}(\omega : \theta(\omega) \in A), \quad A \in \mathcal{G}.$$

2.3 Transition to Probability Measure over Probability Measures

If a distribution is a probability measure then we can consider the distributions themselves as points in a space that we define as follows:

$$\mathcal{M}(\Theta) = \{ \text{ all probability measures on } (\Theta, \mathcal{B}(\Theta))\}$$

Hence, the space is defined as:

$$\mathcal{M}(\Theta) = \{\mu : \mathcal{B}(\Theta) \to [0, 1] \mid \mu \text{ is a probability measure } \}$$

and each element $\mu \in \mathcal{M}(\Theta)$ is a function.

Now, we should define the sigma algebra on the space:

A σ-algebra on $\mathcal{M}(\Theta)$ must be a collection of subsets of $\mathcal{M}(\Theta)$:

$$\mathcal{G} \subseteq \mathcal{P}(\mathcal{M}(\Theta))$$

whereby the elements E satisfy $E \subseteq \mathcal{M}(\Theta)$. We can construct this σ-algebra based on the following evaluation generation:

$$\mathcal{G} = \sigma\left(\left\{e_A^{-1}(B) : A \in \mathcal{B}(\Theta), B \in \mathcal{B}([0, 1])\right\}\right)$$

where:

$$e_A : \mathcal{M}(\Theta) \to [0, 1], \quad e_A(\mu) = \mu(A)$$

taking the function μ, evaluating it on the set A and returning a number $\mu(A)$

and its inverse is defined on the Borel sets $B \subset [0, 1]$

$$e_A^{-1}(B) = \{\mu \in \mathcal{M}(\Theta) : \mu(A) \in B\}$$

That sigma algebra exists (which can be shown by Billingsley) and this construction is native with the "random measure" that we will define:

G is a random measure, defined with the mapping:

$$G : \Omega \to S = \mathcal{M}_{\mathcal{P}}(\Theta)$$

The construction of $\mathcal{G} = \sigma\left(\left\{e_A^{-1}(B) : A \in \mathcal{B}(\Theta), B \in \mathcal{B}([0,1])\right\}\right)$ makes e_A for all A a measurable (A map $f : (S, \mathcal{G}) \to (T, \mathcal{T})$ is measurable iff $f^{-1}(B) \in \mathcal{G}$ for every $B \in \mathcal{T}$)

$$\forall A, \quad e_A : (M, \mathcal{G}) \to ([0,1], \mathcal{B}([0,1])) \text{ is measurable.}$$

So if $G : (\Omega, \mathcal{F}) \to (S, \mathcal{G})$ is a random measure (i.e. measurable), then composition preserves measurability:

$$e_A \circ G : \Omega \to [0,1] \text{ is measurable.}$$

Knowing that $(e_A \circ G) = G(A)$ leading us to claim that $G(A)$ is a real-valued random variable $G(A) : \Omega \to [0,1]$ for every measurable A, which is the crucial observation when switching to the Dirichlet process. Below is a brief comparison table for the random measures:

Table 1: Comparison of Ordinary Probability and Probability over Measures

Ordinary probability	**Probability over measures**
Space Ω	Space $\mathcal{M}(\Theta)$
Outcomes ω	Outcomes = probability measures μ
Events = sets of outcomes	Events = sets of measures
σ-algebra $\mathcal{F}$	σ-algebra $\mathcal{G}$

Symbol	**Lives in**	**Meaning**
ω	Ω	outcome of randomness
A	$\mathcal{B}(\Theta)$	measurable subset of parameter/data space
G	$\Omega \to \mathcal{M}_P(\Theta)$	random probability measure
$G(\omega)$	$\mathcal{B}(\Theta) \to [0,1]$	ordinary probability measure
$G(A)$	$\Omega \to [0,1]$	real-valued random variable
$G(\omega)(A)$	$[0,1]$	a single number

Table 2: Mathematical notation and their meanings.

and the probability measure over probability measures $(\mathcal{M}_P(\Theta), \mathcal{G})$ is defined as:

$$\Pi(B) := \mathbb{P}(G \in B) = \mathbb{P}(\{\omega \in \Omega : G(\omega) \in B\}), \quad B \in \mathcal{G}$$

3 Dirichlet Process

Using the formalism of Ferguson [4] we can define the Dirichlet Process $\mathrm{DP}(\alpha, G_0)$ is a probability law on the space $\mathcal{M}(\Theta)$ of probability measures on Θ, such that for every finite measurable partition $(A_1, \ldots, A_r)$, the random vector of masses $(G(A_1), \ldots, G(A_r))$ has a Dirichlet distribution with param-

eters $(\alpha G_0(A_1), \ldots, \alpha G_0(A_r))$ where G_0 is the base measure and which can be any probability measure on $(\Theta, \mathcal{B}(\Theta)))$. If we are to express the close form for this finite partition, we should consider the common definition for the Dirichlet distribution defined on the simplex $x_i \geq 0, \sum_{i=1}^{K} x_i = 1$: If [8]

$$(X_1, \ldots, X_K) \sim \text{Dirichlet}(\alpha_1, \ldots, \alpha_K), \quad \alpha_i > 0.$$

$$f(x_1, \ldots, x_K) = \frac{\Gamma\left(\sum_{i=1}^{K} \alpha_i\right)}{\prod_{i=1}^{K} \Gamma(\alpha_i)} \prod_{i=1}^{K} x_i^{\alpha_i - 1}$$

Then, in our case

$$(G(A_1), \ldots, G(A_K)) \sim \text{Dirichlet}(\alpha G_0(A_1), \ldots, \alpha G_0(A_K))$$

$$f(g_1, \ldots, g_K) = \frac{\Gamma(\alpha)}{\prod_{i=1}^{K} \Gamma(\alpha G_0(A_i))} \prod_{i=1}^{K} g_i^{\alpha G_0(A_i) - 1}$$

with

- $g_i = G(A_i)$,

- $\sum_i g_i = 1$,

- $\alpha = \sum_i \alpha G_0(A_i)$ (since G_0 is a probability measure).

DP is a law on random measures characterized by Dirichlet finite-partition mass vectors (explanation based on the finite partition is utilized for processes). The following notation defines the random measure G as a sample $G \sim \text{DP}(\alpha, G_0)$ with G is a random element of $\mathcal{M}(\Theta)$, and its law (distribution) is Π_{α, G_0}, a probability measure on $(\mathcal{M}(\Theta), \mathcal{G})$. In measure theoretic notation:

$$\Pi \circ (e_{A_1}, \ldots, e_{A_k})^{-1} = \text{Dirichlet}(\alpha G_0(A_1), \ldots, \alpha G_0(A_k))$$

3.1 Construction of the Random Measure G

We'll adopt the derivation of Sethuraman for the construction of G random measure:

First we should note that for any sampling scheme, the following holds: Let G_0 is the probability measure on $(\Theta, \mathcal{B})$ assigning mass to sets $G_0(B_j) \in [0, 1]$. A random variable $Y_1 : \Omega \to \Theta$ sampled in $Y_1 \sim G_0$ satisfies

$$\mathbb{P}(Y_1 \in B) = G_0(B) \text{ for all measurable } B \subseteq \Theta$$

3.1.1 Sethuraman Method

We will first start by definitions and utilize the Sethuraman's derivation that these definitions are indeed the ones with which the random measures of the Dirichlet process inherently complies.

Defining G by the self-similar decomposition based on the atoms $\theta_1, \theta_2, \theta_3, \ldots \overset{iid}{\sim} G_0$.

$$G(\boldsymbol{V}, \theta; B) = V_1 \delta_{\theta_1}(B) + (1 - V_1)\, G\left(\boldsymbol{V}^*, \theta^*; B\right).$$

Noticing that $(\boldsymbol{V}^*, \theta^*)$ has the same distribution as $(\boldsymbol{V}, \theta)$ and is independent of $(\ V_1, \theta_1\ )$. Thus, we can re-write the previous equation as the following distributional equation for G :

$$G = V_1 \delta_{\theta_1} + (1 - V_1)\, G^{(1)}$$

where $G^{(1)}$ is an independent copy of G (same distribution as G, independent of V_1, θ_1) This formalism can be interpreted as the allocated mass V_1 to the first atom θ_1 and the remaining mass $1-V_1$ is distributed according to a fresh newly generated DP-governed random measure $G^{(1)}$.

Applying the recursion to that construction gives $G^{(1)} = V_2 \delta_{\theta_2} + (1 - V_2)\, G^{(2)}$

$$\begin{aligned} G &= V_1 \delta_{\theta_1} + (1 - V_1) \left[V_2 \delta_{\theta_2} + (1 - V_2)\, G^{(2)} \right] \\ &= V_1 \delta_{\theta_1} + (1 - V_1)\, V_2 \delta_{\theta_2} + (1 - V_1)\,(1 - V_2)\, G^{(2)} \end{aligned}$$

and one further:

$$G = V_1 \delta_{\theta_1} + (1 - V_1)\, V_2 \delta_{\theta_2} + (1 - V_1)\,(1 - V_2)\, V_3 \delta_{\theta_3} + (1 - V_1)\,(1 - V_2)\,(1 - V_3)\, G^{(3)}$$

In closed form expression:

$$G = \sum_{k=1}^{\infty} \left(V_k \prod_{m<k} (1 - V_m) \right) \delta_{\theta_k}$$

giving the final so-called **stick breaking** form:

$$\text{If } \theta_k \overset{iid}{\sim} G_0, \quad V_k \overset{iid}{\sim} \text{Beta}(1, \alpha), \quad \pi_k = V_k \prod_{m<k} (1 - V_m)\,, \text{ then } G = \sum_{k \geq 1} \pi_k \delta_{\theta_k} \sim \text{DP}\left(\alpha, G_0\right)$$

meaning $G(A) = \sum_{k=1}^{\infty} \pi_k \mathbf{1}\left\{\theta_k \in A\right\}$. Now we build the notation on the finite partition of the space and switch to the vectorized notation, whereby the existence and uniqueness proves the DP:

Fixing any finite measurable partition:

$$\left(A_1, \ldots, A_k\right), \quad A_i \in \mathcal{B}, A_i \cap A_j = \varnothing, \bigcup_{i=1}^{k} A_i = \Theta.$$

Defining the mass vector on the simplex:

$$P := \left(G\left(A_1\right), \ldots, G\left(A_k\right)\right) \in \Delta^{k-1}$$

and building the vector based on A_i and thus $G\left(A_i\right) = V_1 \delta_{\theta_1}\left(A_i\right) + (1 - V_1)\, G'\left(A_i\right)$

$$P = V_1 D + (1 - V_1)\, P'$$

where

$$D := (\delta_{\theta_1}(A_1), \ldots, \delta_{\theta_1}(A_k))$$

and $P' \overset{d}{=} P$ independent of (V_1, D). Because θ_1 falls into exactly one partition cell: there exists (due to non-intersecting partition) a unique index J such that $\theta_1 \in A_J$. Hence, $D = e_J$ which is the J-th standard basis vector. Moreover $\mathbb{P}(D = e_j) = \mathbb{P}(\theta_1 \in A_j) = G_0(A_j)$, since $\theta_1 \sim G_0$. We can change the notation since D corresponds to standard unit vector:

$$E \in \{e_1, \ldots, e_k\}, \quad \mathbb{P}(E = e_j) = G_0(A_j)$$

Hence the final expression is

$$P \overset{d}{=} V_1 E + (1 - V_1) P',$$

with

- $V_1 \sim \text{Beta}(1, \alpha)$,

- E one-hot with $\mathbb{P}(E = e_j) = G_0(A_j)$,

- $P' \overset{d}{=} P$ independent of (V_1, E).

The proof relies on existence then uniqueness derivations: if we have $P' \sim \text{Dirichlet}(\alpha G_0(A_1), \ldots, \alpha G_0(A_k))$, and the random vector $\widetilde{P} := V_1 E + (1 - V_1) P'$, formed with the definitions above must have the same Dirichlet distributions (existence):

$$\widetilde{P} \sim \text{Dirichlet}(\alpha G_0(A_1), \ldots, \alpha G_0(A_k))$$

The uniqueness is explained again by Sethuraman stating that on a simplex, $P \overset{d}{=} V_1 E + (1 - V_1) P'$ form can have no more than 1 distribution (i.e. if P satisfies this recursion, it is the only one in the simplex). Hence, the existence and uniqueness leads to the identification of the construction based on discrete atoms sampled independently from the base measure G_0.

3.2 The Relation of Dirichlet Process to Chinese Restaurant Process

After we construct G with the formulation above $G := \sum_{k=1}^{\infty} \pi_k \delta_{\theta_k}$, being equivalent to $G \sim \text{DP}(\alpha, G_0)$, which is now known. Hence, we sample the parameter on that measure $\theta_i \mid G \overset{\text{i.i.d.}}{\sim} G$. Thus,

$$\mathbb{P}(\theta_i \in A \mid G) = G(A) \quad \text{for every measurable } A \subseteq \Theta.$$

and in our partition $(A_1, \ldots, A_k)$:

$$\mathbb{P}(\theta_i \in A_j \mid G) = G(A_j)$$

Now, we apply a reparameterization with label parameter based on our partition: $Z_i := j$ iff $\theta_i \in A_j$. So each θ_i induces exactly one label $Z_i \in \{1, \ldots, k\}$. Hence, $\mathbb{P}(Z_i = j \mid G) = \mathbb{P}(\theta_i \in A_j \mid G) = G(A_j)$. So, we can infer that conditional on G, the labels Z_i are i.i.d. Categorical with probabilities $(G(A_1), \ldots, G(A_k))$.

Hence, the likelihood of the parameter history 1:n given a random measure is expressed as

$$\mathbb{P}\left(Z_{1:n} \mid G\right) = \mathbb{P}\left(\theta_{1:n} \mid G\right) = \prod_{i=1}^{n} G\left(A_{Z_i}\right) = \prod_{j=1}^{k} G\left(A_j\right)^{n_j}$$

where $n_j := \sum_{i=1}^{n} 1\left\{\theta_i \in A_j\right\}$. Hence the dependence of the data $\theta_{1:n}$ on G_A is through the counts $n_1, \ldots, n_k$, which is similar to the idea of sufficient statistics. Then, expressing the parameter history likelihood is equivalent to expressing the multinomial likelihood of the parameter counts n:

$$n \mid G_A \sim \text{Multinomial}\left(n; G\left(A_1\right), \ldots, G\left(A_k\right)\right)$$

This will help us when we try to get the posterior:

For a prior $G_{\mathcal{A}} \sim \text{Dirichlet}\left(\alpha G_0\left(A_1\right), \ldots, \alpha G_0\left(A_k\right)\right) : \pi(p) \propto \prod_{j=1}^{k} p_j^{\alpha G_0(A_j)-1}$ and defining $p_j = G\left(A_j\right)$ the likelihood becomes:

$$\mathbb{P}\left(Z_{1:n} \mid p\right) = \prod_{i=1}^{n} p_{Z_i} = \prod_{j=1}^{k} p_j^{n_j} \quad \text{(up to the multinomial coefficient)}.$$

or equivalently with different notation $L\left(p; \theta_{1:n}\right) \propto \prod_{j=1}^{k} p_j^n$. Posterior density is:

$$\pi\left(p \mid \theta_{1:n}\right) \propto L\left(p; \theta_{1:n}\right) \pi(p) \propto \prod_{j=1}^{k} p_j^{n_j} \prod_{j=1}^{k} p_j^{\alpha G_0(A_j)-1} = \prod_{j=1}^{k} p_j^{\alpha G_0(A_j)+n_j-1}.$$

Hence, for finite partitions:

$$G_{\mathcal{A}} \mid \theta_{1:n} \sim \text{Dirichlet}\left(\alpha G_0\left(A_1\right) + n_1, \ldots, \alpha G_0\left(A_k\right) + n_k\right)$$

In order to derive the distribution law on the random measure, we should have the following common form $G \mid \theta_{1:n} \sim \text{DP}\left(\alpha + n, \tilde{G}_0\right)$, for which we need to find the expression for the $\tilde{G}_0$: The number addition leads us to think whether Dirac measures can be used to alter the base measure for this posterior. δ_{θ_i} is the measure that puts mass 1 at θ_i:

$$\delta_{\theta_i}(A) := \mathbf{1}\left\{\theta_i \in A\right\}.$$

Defining $S := \sum_{i=1}^{n} \delta_{\theta_i}.$, for any measurable set A, $S(A) = \sum_{i=1}^{n} \delta_{\theta_i}(A) = \sum_{i=1}^{n} \mathbf{1}\left\{\theta_i \in A\right\}$. Hence this S measure is the one that counts how many parameters lie (i.e. n_j in the argument set A_j

$$S\left(A_j\right) = n_j.$$

Defining $M := \alpha G_0 + S$, which is a measure (non-normalized).

$$M\left(A_j\right) = \alpha G_0\left(A_j\right) + S\left(A_j\right) = \alpha G_0\left(A_j\right) + n_j.$$

The total mass $M(\Theta) = \alpha G_0(\Theta) + S(\Theta) = \alpha + n$. Since our goal is to obtain a probability measure, we need to normalize it:

$$\tilde{G}_0 := \frac{M}{M(\Theta)} = \frac{\alpha G_0 + \sum_{i=1}^{n} \delta_{\theta_i}}{\alpha + n}$$

In many cases, G is the formalism and underlying mechanism for the parameter observation, but we need to answer the question for any measurable set $A \in \mathcal{B}(\Theta)$

$$\mathbb{P}\left(\theta_{n+1} \in A \mid \theta_{1:n}\right)$$

To be able to find an expression for that we can apply the following procedure:

Conditioning on $\theta_{1:n}$ and averaging over the posterior uncertainty in G (effectively marginalization) gives:

$$\mathbb{P}\left(\theta_{n+1} \in A \mid \theta_{1:n}\right) = \mathbb{E}\left[\mathbb{P}\left(\theta_{n+1} \in A \mid G, \theta_{1:n}\right) \mid \theta_{1:n}\right]$$

Here, we know that once the random measure is fixed (by conditioning) all the parameter samples are iid. Hence, the condition on G removes the need for the condition on the history:

$$\mathbb{P}\left(\theta_{n+1} \in A \mid G, \theta_{1:n}\right) = \mathbb{P}\left(\theta_{n+1} \in A \mid G\right) = G(A)$$

$$\mathbb{P}\left(\theta_{n+1} \in A \mid \theta_{1:n}\right) = \mathbb{E}\left[G(A) \mid \theta_{1:n}\right]$$

We can compute $\mathbb{E}\left[G(A) \mid \theta_{1:n}\right]$ using the DP posterior: $G \mid \theta_{1:n} \sim \mathrm{DP}\left(\alpha + n, \tilde{G}_0\right)$. Following Ferguson's partition with the 2 -set partition (A, A^c)

$$\left(G(A), G\left(A^c\right)\right) \mid \theta_{1:n} \sim \text{Dirichlet}\left((\alpha + n)\tilde{G}_0(A), (\alpha + n)\tilde{G}_0\left(A^c\right)\right)$$

It is noted that Dirichlet on two coordinates reduces to a Beta distribution:

$$G(A) \mid \theta_{1:n} \sim \text{Beta}\left((\alpha + n)\tilde{G}_0(A), (\alpha + n)\tilde{G}_0\left(A^c\right)\right)$$

Knowing that Beta distribution $\text{Beta}(u, v)$ has the mean of $u/(u + v)$ [9].

Here $u + v = (\alpha + n)\left(\tilde{G}_0(A) + \tilde{G}_0\left(A^c\right)\right) = (\alpha + n)$. Then,

$$\mathbb{E}\left[G(A) \mid \theta_{1:n}\right] = \tilde{G}_0(A)$$

$$\mathbb{P}\left(\theta_{n+1} \in A \mid \theta_{1:n}\right) = \tilde{G}_0(A)$$

Now we need to write the expression:

$$\tilde{G}_0 = \frac{\alpha G_0 + \sum_{i=1}^{n} \delta_{\theta_i}}{\alpha + n}$$

$$\tilde{G}_0(A) = \frac{\alpha G_0(A) + \sum_{i=1}^{n} \delta_{\theta_i}(A)}{\alpha + n}$$

Knowing that $\delta_{\theta_i}(A) = \mathbf{1}\{\theta_i \in A\}$

$$\tilde{G}_0(A) = \frac{\alpha G_0(A) + \sum_{i=1}^{n} 1\{\theta_i \in A\}}{\alpha + n}$$

$$\mathbb{P}(\theta_{n+1} \in A \mid \theta_{1:n}) = \frac{\alpha}{\alpha + n} G_0(A) + \frac{1}{\alpha + n} \sum_{i=1}^{n} \mathbf{1}\{\theta_i \in A\}$$

For the definition $\hat{G}_n := \frac{1}{n} \sum_{i=1}^{n} \delta_\theta$, then the predictive is a convex combination:

$$\mathbb{P}(\theta_{n+1} \in \cdot \mid \theta_{1:n}) = \frac{\alpha}{\alpha + n} G_0(\cdot) + \frac{n}{\alpha + n} \hat{G}_n(\cdot)$$

which can be interpreted as With probability $\alpha/(\alpha + n)$ generate a new value from G_0 and with probability $n/(\alpha + n)$ copy a previous value

This formalism can be turned into a simple algorithm for the sampling: we generate $U \sim \text{Bernoulli}\left(\frac{\alpha}{\alpha+n}\right)$ and If $U = 1$ we sample a new value should be drawn from the base distribution $\theta_{n+1} \sim G_0$ (implying new atoms are created) and If $U = 0$ we copy an old value by choosing an index $i \in \{1, \ldots, n\}$ uniformly ($\mathbb{P}(i = j) = \frac{1}{n}$) and assigning $\theta_{n+1} := \theta_i$

An equivalent expression makes use of $\sum_{i=1}^{n} \delta_{\theta_i} = \sum_{k=1}^{K} n_k \delta_{\theta_k^*}$ for $\theta_1^*, \ldots, \theta_K^*$ distinct values:

$$\theta_{n+1} | \theta_{1:n} \sim \sum_{k=1}^{K} \frac{n_k}{\alpha + n} \delta_{\theta_k^*} + \frac{\alpha}{\alpha + n} G_0$$

Another perspective is that sampling from joint distribution $p(\theta_1, \ldots, \theta_n)$ reduces to predictive rule and sequential sampling we have derived above due to the following decomposition:

$$p(\theta_1, \ldots, \theta_n) = p(\theta_1) \prod_{i=2}^{n} p(\theta_i \mid \theta_{1:i-1}).$$

These formulae are in conformity with what Neal [6] proposes.

4 Dirichlet Process Mixture Models (DPMM) in Clustering Perspective

Neal [6] also mentions the connection of Dirichlet process and the mixture models describing the data generation with the following hierarchy

$$\begin{aligned} x_i \mid \theta_i &\sim F(\theta_i) \\ \theta_i \mid G &\sim G \\ G &\sim \text{DP}(G_0, \alpha). \end{aligned}$$

So $F(B \mid \theta)$ means $\mathbb{P}(X \in B \mid \theta)$ for $B \in \mathcal{A}$.The mixture induced by G is then

$$P_G(B) := \int_\Theta F(B \mid \theta) G(d\theta), \quad B \in \mathcal{A}$$

where $G(d\theta)$ meaning "integrate with respect to the measure G". Knowing the construction formulation

$$G = \sum_{k=1}^{\infty} \pi_k \delta_{\phi_k}$$

for atoms $\phi_k \in \Theta$ and weights $\pi_k \geq 0, \sum_k \pi_k = 1$

$$P_G(B) = \int F(B \mid \theta) G(d\theta) = \sum_{k=1}^{\infty} \pi_k F\left(B \mid \phi_k\right),$$

Hence, Dirichlet process inherently yields an infinite mixture.

If we look at the distribution as the model to generate the data, we can be more interested in $P\left(x_{n+1} \in B \mid \theta_{1:n}\right)$. Knowing that:

$$\theta_{n+1} \,\Big|\, \theta_{1:n} \sim \frac{1}{\alpha + n} \sum_{i=1}^{n} \delta_{\theta_i} + \frac{\alpha}{\alpha + n} G_0.$$

$$P\left(d\theta \mid \theta_{1:n}\right) = \tilde{G}_0(d\theta) = \frac{\alpha}{\alpha + n} G_0(d\theta) + \frac{1}{\alpha + n} \sum_{i=1}^{n} \delta_{\theta_i}(d\theta).$$

where $P\left(d\theta \mid \theta_{1:n}\right)$ satisfies $P\left(\theta_{n+1} \in A \mid \theta_{1:n}\right) = \int_A P\left(d\theta \mid \theta_{1:n}\right)$. Hence, our formula can be expressed as:

$$P\left(x_{n+1} \in B \mid \theta_{1:n}\right) = \int_\Theta F(B \mid \theta) P\left(d\theta \mid \theta_{1:n}\right)$$

$$\int_\Theta F(B \mid \theta) \left[\frac{\alpha}{\alpha + n} G_0(d\theta) + \frac{1}{\alpha + n} \sum_{i=1}^{n} \delta_{\theta_i}(d\theta) \right].$$

Knowing that $\int_\Theta f(\theta) \delta_{\theta_i}(d\theta) = f\left(\theta_i\right)$,

$$\frac{1}{\alpha + n} \sum_{i=1}^{n} \int_\Theta F(B \mid \theta) \delta_{\theta_i}(d\theta) = \frac{1}{\alpha + n} \sum_{i=1}^{n} F\left(B \mid \theta_i\right).$$

$$P\left(x_{n+1} \in B \mid \theta_{1:n}\right) = \frac{\alpha}{\alpha + n} \int_\Theta F(B \mid \theta) G_0(d\theta) + \frac{1}{\alpha + n} \sum_{i=1}^{n} F\left(B \mid \theta_i\right).$$

or equivalently:

$$P\left(x_{n+1} \in B \mid \theta_{1:n}\right) = \frac{\alpha}{\alpha+n}\int_{\Theta} F(B \mid \theta) G_0(d\theta) + \sum_{k=1}^{K} \frac{n_k}{\alpha+n} F\left(B \mid \theta_k^*\right).$$

4.1 Clustering

Let's use the reparameterization similar to what we have discussed before: Given $\theta_{1:n}$, we can construct $(z_{1:n}, \phi_{1:K_n})$ where $\phi_1, \ldots, \phi_{K_n}$ are the distinct values appearing in $\{\theta_1, \ldots, \theta_n\}$

$$\theta_i = \phi_{z_i}, \quad z_i \in \{1, \ldots, K_n\}$$

and due to the one-to-one relation, given $(z_{1:n}, \phi_{1:K_n})$, we can reconstruct $\theta_i = \phi_{z_i}$. For instance for predictive $\theta_{n+1} \,\Big|\, \theta_{1:n} \sim \frac{\alpha}{\alpha+n} G_0 + \sum_{k=1}^{K_n} \frac{n_k}{\alpha+n} \delta_{\phi k}$.

- $z_{n+1} = k$ if $\theta_{n+1} = \phi_k$ for an existing atom,

- $z_{n+1} = K_n + 1$ if θ_{n+1} is a new draw (a new atom). Then we have the exact Chinese Restaurant Process rule

$$\mathbb{P}\left(z_{n+1} = k \mid z_{1:n}\right) = \frac{n_k}{\alpha+n}, \quad k = 1, \ldots, K_n$$
$$\mathbb{P}\left(z_{n+1} = K_n + 1 \mid z_{1:n}\right) = \frac{\alpha}{\alpha+n}.$$

Hence, the claim becomes the following: A Dirichlet Process mixture is a Bayesian mixture model with infinitely many potential components. During collapsed inference, we do not explicitly represent cluster parameters (non-parametric model).Instead, we integrate them out analytically using conjugacy and sample only cluster assignments. After clustering is done, if we want, we can sample the cluster parameters from their posterior given the data assigned to each cluster. Now we will lay the details how this approach is possible and how to obtain its steps clearly:

For clustering, we can apply Gibbs approach and look for $p\left(z_i = \kappa \mid z_{-i}, x_{1:N}\right)$ where z_{-i} means all assignments except z_i and cluster κ could be an existing cluster (already used by some $z_j, j \neq i$) or a new one. Now we should apply statistical manipulations and make use of the model properties to obtain components that we are familiar with:

$$p\left(z_i = \kappa \mid z_{-i}, x\right) = \frac{p\left(x \mid z_i = \kappa, z_{-i}\right) p\left(z_i = \kappa \mid z_{-i}\right)}{p\left(x \mid z_{-i}\right)}.$$

and the denominator does not depend on the value κ. Thus, $p\left(z_i = \kappa \mid z_{-i}, x\right) \propto p\left(x \mid z_i = \kappa, z_{-i}\right) p\left(z_i = \kappa \mid z_{-i}\right)$. Knowing that:

$$p\left(x_i, x_{-i} \mid \text{ condition }\right) = p\left(x_i \mid x_{-i}, \text{ condition }\right) p\left(x_{-i} \mid \text{ condition }\right)$$

$$p\left(x \mid z_i = \kappa, z_{-i}\right) = p\left(x_i \mid z_i = \kappa, z_{-i}, x_{-i}\right) p\left(x_{-i} \mid z_{-i}\right)$$

However, in our modeling $p\left(x_i \mid x_{-i}, z_i = \kappa, z_{-i}\right)$ does not depend on κ since each x_j depends only on its own cluster label z_j, not on other labels. So changing z_i only changes the distribution of x_i, not the already-existing points x_{-i} whose labels z_{-i} are fixed. Hence, this simplification yields

$$p\left(x_{-i} \mid z_i = \kappa, z_{-i}\right) = p\left(x_{-i} \mid z_{-i}\right)$$

which becomes independent of κ. Finally, we are left with the following κ dependent terms: $p\left(z_i = \kappa \mid z_{-i}, x\right) \propto p\left(x_i \mid x_{-i}, z_i = \kappa, z_{-i}\right) p\left(z_i = \kappa \mid z_{-i}\right)$.

We know that $p\left(z_i = \kappa \mid z_{-i}\right)$ is expressed by the conditional labeling we have derived above,separating into two scenarios of either joining to an existing cluster $p\left(z_i = k \mid z_{-i}\right) = \frac{n_{-i,k}}{\alpha+(n-1)}$, or having a new cluster $p\left(z_i = \text{ new } \mid z_{-i}\right) = \frac{\alpha}{\alpha+(n-1)}$.

We should then focus on the $p\left(x_i \mid z_i = \kappa, z_{-i}, x_{-i}\right)$ term:

When it is one of the existing labels

$p\left(x_i \mid z_i = k, z_{-i}, x_{-i}\right)$ emerges and it can be expressed as:

$$p\left(x_i \mid z_i = k, z_{-i}, x_{-i}\right) = \int p\left(x_i, \phi_k \mid z_i = k, z_{-i}, x_{-i}\right) d\phi_k$$

$$p\left(x_i \mid z_i = k, z_{-i}, x_{-i}\right) = \int p\left(x_i \mid \phi_k, z_i = k, z_{-i}, x_{-i}\right) p\left(\phi_k \mid z_i = k, z_{-i}, x_{-i}\right) d\phi_k$$

Here we use the conditional independence: given label parameters, the likelihood of x_i is independent of other labels or data points.

$$p\left(x_i \mid \phi_k, z_i = k, z_{-i}, x_{-i}\right) = p\left(x_i \mid \phi_k, z_i = k\right) = p\left(x_i \mid \phi_k\right)$$

For the other term $p\left(\phi_k \mid z_i = k, z_{-i}, x_{-i}\right)$ we should simplify it further. Knowing that:

$$p(x \mid z, \phi) = \prod_{j=1}^{n} p\left(x_j \mid z_j, \phi\right) = \prod_{k} \prod_{j:z_j=k} F\left(x_j \mid \phi_k\right).$$

$$p\left(x \mid z, \phi_k, \phi_{-k}\right) = \left(\prod_{j:z_j=k} F\left(x_j \mid \phi_k\right)\right) \times \underbrace{\left(\prod_{\ell \neq k} \prod_{j:z_j=\ell} F\left(x_j \mid \phi_\ell\right)\right)}_{\text{does not depend on } \phi_k}.$$

Hence, the posterior $p\left(\phi_k \mid z, x\right) \propto p\left(\phi_k\right) p\left(x \mid z, \phi_k, \phi_{-k}\right)$:

$$p\left(\phi_k \mid z, x\right) \propto p\left(\phi_k\right) \prod_{j:z_j=k} F\left(x_j \mid \phi_k\right).$$

Thus,

$$p\left(\phi_k \mid z_i = k, z_{-i}, x_{-i}\right) = p\left(\phi_k \mid x_{-i,k}\right)$$

where $x_{-i,k} = \{x_j : j \neq i, z_j = k\}$: points in cluster k excluding i. Hence, the final expression becomes

$$p\left(x_i \mid z_i = k, z_{-i}, x_{-i}\right) = \int p\left(x_i \mid \phi_k\right) p\left(\phi_k \mid x_{-i,k}\right) d\phi_k = p\left(x_i \mid x_{-i,k}\right).$$

When it creates a new index

$$p\left(x_i \mid z_i = \text{ new }, z_{-i}, x_{-i}\right) = \int p\left(x_i \mid \phi_{\text{new}}, z_i = \text{ new }, z_{-i}, x_{-i}\right) dG_0((\phi_{\text{new}}))$$

and knowing that $p\left(x_i \mid \phi_{\text{new}}, z_i = \text{ new }, z_{-i}, x_{-i}\right) = p\left(x_i \mid \phi_{\text{new}}\right) = F\left(x_i \mid \phi_{\text{new}}\right)$

$$p\left(x_i \mid z_i = \text{ new }, z_{-i}, x_{-i}\right) = \int F\left(x_i \mid \phi\right) G_0(d\phi).$$

or as a short-hand notation $p\left(x_i \mid G_0\right) := \int F\left(x_i \mid \phi\right) G_0(d\phi)$

If G_0 has a density $g_0(\phi)$ with respect to Lebesgue measure, then the same thing can be written as the familiar density integral

$$\int F\left(x_i \mid \phi\right) g_0(\phi) d\phi$$

Hence as a summary, the conditional parameter update is described as ,

Existing cluster k :

$$p\left(z_i = k \mid z_{-i}, x\right) \propto n_{-i,k} \cdot \underbrace{p\left(x_i \mid x_{-i,k}\right)}_{\text{posterior predictive}}$$

New cluster:

$$p\left(z_i = \text{ new } \mid z_{-i}, x\right) \propto \alpha \cdot \underbrace{\int F\left(x_i \mid \phi\right) G_0(d\phi)}_{\text{prior predictive}}$$

which eventually complies with the Neal's expression.

4.2 Implementation Details of the Clustering

For the data $x_i \in \mathbb{R}^d, i = 1, \ldots, N$ the Gibbs method needs an initialization. Hence, we can run k-means with K_{init} or randomly assign the data points. For the integrals we have discussed to have analytical solutions, conjugates should be utilized [10]:

For gaussian F likelihood model $x \mid (\mu, \Sigma) \sim \mathcal{N}(\mu, \Sigma)$ we choose Normal-Inverse-Wishart for the parameter prior (in our case base Measure G_0)

$$(\mu, \Sigma) \sim \text{NIW}\left(\mu_0, \kappa_0, \nu_0, \Lambda_0\right)$$

We will need sufficient statistics sum ($s_k = \sum_{j:z,=k} x_j$), scatter($S_k = \sum_{j:z_j=k} x_j x_j^\top$), empirical mean ($\bar{x}_k = s_k/n_k$) and centered scatter ($C_k = \sum \left(x - \bar{x}_k\right)\left(x - \bar{x}_k\right)^\top = S_k - n_k \bar{x}_k \bar{x}_k^\top$). The posterior NIW has the following parameters:

$$\kappa_k = \kappa_0 + n_k$$
$$\nu_k = \nu_0 + n_k$$
$$\mu_k = \frac{\kappa_0 \mu_0 + n_k \bar{x}_k}{\kappa_0 + n_k}$$
$$\Lambda_k = \Lambda_0 + C_k + \frac{\kappa_0 n_k}{\kappa_0 + n_k} (\bar{x}_k - \mu_0) (\bar{x}_k - \mu_0)^\top$$

which come from completing the square in the Gaussian likelihood. As shown in [12] and [11] the predictive densities have the following distributions:

$$p(x \mid G_0) = t_{\nu_0 - d + 1} \left(x; \mu_0, \frac{\kappa_0 + 1}{\kappa_0 (\nu_0 - d + 1)} \Lambda_0 \right)$$

$$p(x_i \mid x_{-i,k}) = t_{\nu_k - d + 1} \left(x_i; \mu_k, \frac{\kappa_k + 1}{\kappa_k (\nu_k - d + 1)} \Lambda_k \right)$$

which follow Student-t distributions. Thus, Gibbs sampler explores the posterior distribution over partitions induced by a Dirichlet Process mixture model. At each iteration, it removes one data point, reassigns it using CRP prior (indicating the cluster sizes as a way of interpretation) and predictive likelihoods. By doing so it possibly creates or deletes the clusters dynamically. This dynamic adaptation determines the cluster counts, cluster shapes(implicit) and the assignments

4.3 Interpretations

Knowing that we have derived:

$$\mathbb{P}(x_{n+1} \in B \mid \theta_{1:n}) = \frac{\alpha}{\alpha + n} \int_\Theta F(B \mid \theta) G_0(d\theta) + \sum_{k=1}^{K} \frac{n_k}{\alpha + n} F(B \mid \phi_k^*)$$

with cluster parameters ϕ_k and with cluster labels $z_i \in \{1, 2, \ldots\}$. We can obtain this generation perspective with no explicit parameter knowledge (conforming with the notion of non-parametric approach). The generation is modeled by the distribution: $p(x_{n+1} \mid z_{1:n}, x_{1:n})$:

$$p(x_{n+1} \mid z_{1:n}, x_{1:n}) = \sum_{k=1}^{K} \mathbb{P}(z_{n+1} = k \mid z_{1:n}) p(x_{n+1} \mid z_{n+1} = k, z_{1:n}, x_{1:n}) + \mathbb{P}(\text{ new } \mid z_{1:n}) p(x_{n+1} \mid \text{ new }, z_{1:n}, x_{1:n}).$$

where

$$\mathbb{P}(z_{n+1} = k \mid z_{1:n}) = \frac{n_k}{\alpha + n}, \quad \mathbb{P}(z_{n+1} = \text{ new } \mid z_{1:n}) = \frac{\alpha}{\alpha + n}.$$

Knowing that we have also derived

$$p(x_{n+1} \mid z_{n+1} = k, z_{1:n}, x_{1:n}) = \int_\Theta F(x_{n+1} \mid \phi) p(\phi \mid x_k) d\phi$$

$$p(x_{n+1} \mid z_{n+1} = \text{ new }, z_{1:n}, x_{1:n}) = \int_\Theta F(x_{n+1} \mid \phi) G_0(d\phi)$$

$$p\left(x_{n+1} \mid z_{1:n}, x_{1:n}\right) = \sum_{k=1}^{K} \frac{n_k}{\alpha + n} \underbrace{\int F\left(x_{n+1} \mid \phi\right) p\left(\phi \mid x_k\right) d\phi}_{\text{cluster posterior predictive}} + \frac{\alpha}{\alpha + n} \underbrace{\int F\left(x_{n+1} \mid \phi\right) G_0(d\phi)}_{\text{prior predictive}}$$

For a model designed with the conjugacy: $F(x \mid \mu, \Sigma) = \mathcal{N}(x \mid \mu, \Sigma)$, and the base distribution $G_0(\mu, \Sigma) = \text{NIW}\left(\mu_0, \kappa_0, \nu_0, \Lambda_0\right)$, for which we know that

$$\int \mathcal{N}(x \mid \mu, \Sigma) G_0(d\mu d\Sigma) \text{ is a multivariate Student- } t \text{ (the prior predictive)}$$

$$\int \mathcal{N}(x \mid \mu, \Sigma) p\left(\mu, \Sigma \mid x_k\right) d\mu d\Sigma \text{ is also a multivariate Student- } t \text{ (the posterior predictive),}$$

with the posterior parameters defined in the previous section. As we can see no explicit ϕ_k is needed because the Student- t is already the result of integrating (μ, Σ) out. This fact clearly demonstrates that the DP mixture defines a valid generative model for which no cluster parameters need to be sampled (nonparametric in inference)

The overall Gibbs sampling and Predictive Generation from the DP Mixture algorithms are depicted below:

Algorithm 1 Collapsed Gibbs sampler for a Dirichlet Process Gaussian Mixture (NIW-conjugate)

Data: $x_{1:N}$ with $x_i \in \mathbb{R}^d$. **Likelihood:** $x \mid (\mu, \Sigma) \sim \mathcal{N}(\mu, \Sigma)$. **Base prior (conjugate):** $(\mu, \Sigma) \sim \mathrm{NIW}(\mu_0, \kappa_0, \nu_0, \Lambda_0)$. **DP concentration:** $\alpha > 0$.

Sufficient statistics (per cluster k): count: n_k sum: $s_k = \sum_{i:z_i=k} x_i$ scatter: $S_k = \sum_{i:z_i=k} x_i x_i^\top$

Initialization: Choose initial labels $z_{1:N}$ (e.g., random; all-in-one; or k-means with K_{init}). Compute initial $\{n_k, s_k, S_k\}_{k=1}^K$ from $z_{1:N}$.

for $t = 1, 2, \ldots, T$ **do** `// Gibbs sweeps`

for $i = 1, 2, \ldots, N$ **do** `// Resample each assignment`

`// (1) Remove` x_i `from its current cluster`

Let $k_{\text{old}} \leftarrow z_i$. $n_{k_{\text{old}}} \leftarrow n_{k_{\text{old}}} - 1$ $s_{k_{\text{old}}} \leftarrow s_{k_{\text{old}}} - x_i$ $S_{k_{\text{old}}} \leftarrow S_{k_{\text{old}}} - x_i x_i^\top$ **if** $n_{k_{old}} = 0$ **then**

delete cluster k_{old} and relabel indices to keep clusters $1, \ldots, K$

`// (2) Compute unnormalized weights for existing clusters`

for $k = 1, 2, \ldots, K$ **do**

Define centered scatter:

$$C_k \;=\; S_k - \frac{1}{n_k} s_k s_k^\top.$$

NIW posterior parameters for cluster k:

$$\kappa_k = \kappa_0 + n_k, \qquad \nu_k = \nu_0 + n_k,$$

$$\mu_k = \frac{\kappa_0 \mu_0 + s_k}{\kappa_0 + n_k},$$

$$\Lambda_k = \Lambda_0 + C_k + \frac{\kappa_0 n_k}{\kappa_0 + n_k} (\bar{x}_k - \mu_0)(\bar{x}_k - \mu_0)^\top, \quad \bar{x}_k = \frac{s_k}{n_k}.$$

`// Posterior predictive is multivariate Student-`t

Let $\mathrm{Pred}_k(x)$ denote:

$$x \mid x_{-i,k} \sim t_{\nu_k - d + 1}\left(\mu_k,\; \frac{\kappa_k + 1}{\kappa_k(\nu_k - d + 1)} \Lambda_k\right).$$

Set weight:

$$w_k \propto n_{-i,k} p\left(x_i \mid x_{-i,k}\right).$$

`// (3) New-cluster weight (prior predictive under NIW)`

Define the prior-predictive $\mathrm{Pred}_0(x)$ as:

$$x \sim t_{\nu_0 - d + 1}\left(\mu_0,\; \frac{\kappa_0 + 1}{\kappa_0(\nu_0 - d + 1)} \Lambda_0\right).$$

Set:

$$w_{\text{new}} \;\propto \alpha p\left(x_i \mid G_0\right)$$

`// (4) Normalize and sample new assignment`

Normalize $\{w_1, \ldots, w_K, w_{\text{new}}\}$ to probabilities. Sample z_i from this categorical distribution.

`// (5) Add` x_i `into its (possibly new) cluster`

if *z_i is an existing cluster $k \in \{1, \ldots, K\}$* **then**

$n_k \leftarrow n_k + 1$ $s_k \leftarrow s_k + x_i$ $S_k \leftarrow S_k + x_i x_i^\top$

else

`// Create new cluster` $K{+}1$

$K \leftarrow K + 1$ $z_i \leftarrow K$ $n_K \leftarrow 1$ $s_K \leftarrow x_i$ $S_K \leftarrow x_i x_i^\top$

Output: MCMC samples $\{z_{1:N}^{(t)}\}_{t=1}^T$, number of clusters $K^{(t)}$. Optionally (post-processing): co-clustering matrix or a representative partition (e.g., Dahl's method).

Algorithm 2 Posterior Predictive Sampling from a DP Mixture

Input: Observed data $x_{1:n}$ with cluster assignments $z_{1:n}$;
cluster counts $\{n_k\}_{k=1}^{K}$;
DP concentration α;
base measure G_0;
likelihood family $F(\cdot \mid \theta)$
Output: Sampled new data point x_{n+1}

Sample a label:

$$z_{n+1} \sim \text{Categorical}\left(\frac{n_1}{\alpha+n}, \ldots, \frac{n_K}{\alpha+n}, \frac{\alpha}{\alpha+n}\right).$$

if $z_{n+1} = k \leq K$ **then**
    Sample from the cluster posterior predictive:

$$x_{n+1} \sim p(x \mid x_k),$$

    where $p(x \mid x_k)$ is the student-t distribution obtained by integrating θ under $p(\theta \mid x_k)$.
else
    Sample from the prior predictive:

$$x_{n+1} \sim p(x \mid G_0),$$

    where $p(x \mid G_0) = \int F(x \mid \theta)\, G_0(d\theta)$ yielding closed-form student-t

4.4 Implementation Details in Practice

As opposed to parameter estimation, Bayesian inference yields the **distribution** as the output. To be able to perform experiments, we need to make a decision on the clustering of the data points, under the DP-governed mixture. Gibbs sampling, whose steps are explicitly mentioned in previous sections provide samples ($z^{(1)}, \ldots, z^{(T)}$ after burn-in) coming from the distribution we are looking for. However the intermediary label mixing requires us develop a systematic way to reach a verdict for the clustering and make this question well-posed. For the clustering to be freed of nominal labels and to build a **canonical** object for clustering purposes, we can use **Dahl's** approach [13].

The association matrix is defined for co-clustering purposes.

$$A_{ij}^{(t)} = \mathbf{1}\left\{z_i^{(t)} = z_j^{(t)}\right\}$$

It encodes exactly the equivalence relation of "...is in the same cluster as...". Furthermore, it is label-invariant meaning that relabeling clusters does not change $A^{(t)}$. Defining similarity as:

$$S_{ij} = \mathbb{E}\left[A_{ij} \mid x\right] = \Pr(i \text{ and } j \text{ are in the same cluster } \mid x)$$

we can build it empirically as:

$$S_{ij} \approx \frac{1}{T}\sum_{t=1}^{T} A_{ij}^{(t)}.$$

which acts as the empirical version of the mean $S = \mathbb{E}[A \mid x]$.

Hence, for the true clustering object A, MMSE estimation $\mathbb{E}\left[\left\|A^{(t)} - A\right\|_F^2 \mid x\right]$ corresponds to the minimization of the following distance:

$$D(t) = \sum_{i<j} \left(A_{ij}^{(t)} - S_{ij} \right)^2$$

with the the solution

$$A^{(t^*)} = \arg\min_{A^{(t)}} \left\| A^{(t)} - S \right\|_F^2$$

and thus the choice is $z^* = z^{(t^*)}$, meaning to choose the sample that minimizes the MSE error for the clustering (i.e. $t^* = \arg\min_t D(t)$).

4.4.1 Determining the Concentration Hyper-parameter α

By the so-called chain rule of the probability, we know $p(z \mid \alpha)$ as $p\left(z_{1:N} \mid \alpha\right) = \prod_{i=1}^{N} p\left(z_i \mid z_{1:i-1}, \alpha\right)$. The α hyper-parameter determines how likely the model is prone to create a new cluster when the data comes (direct consequence of the CRP formulation). If α is large, the model tends to have larger number of cluster prediction and vice versa. Thus, it plays an important role in the model behavior and cluster estimation. We can adjust this parameter based on the data properties, using common statistical tools. We are to learn the posterior $p(\alpha \mid z)$, we need to define the prior on α too Hence, Escobar and West [14] address this problem as follows:

They assign a prior $\alpha \sim \text{Gamma}(a, b)$. Instead of directly solving the posterior update, they define an auxiliary parameter η and they apply the following:

Given current α, they sample $\eta \mid \alpha, N \sim \text{Beta}(\alpha + 1, N)$ and compute the mixture weights:

$$\pi(\eta) = \frac{a + K - 1}{a + K - 1 + N(b - \log \eta)}.$$

with a Bernoulli variable: $s \sim \text{Bernoulli}(\pi(\eta))$ posterior sampling is based on the following rule:

$$\alpha \mid \eta, K \sim \begin{cases} \text{Gamma}(a + K, b - \log \eta) & \text{if } s = 1, \\ \text{Gamma}(a + K - 1, b - \log \eta) & \text{if } s = 0. \end{cases}$$

as described by the paper.

Therefore, after one sweep of Gibbs sampling of vector z, before the next iteration, we can update the α concentration parameter based on the rule above, which leads us to a simple alternating steps within Gibbs sampling.

5 Experiments

After laying these foundations, the experiments are performed to verify whether Dirichlet process can be used indeed for clustering purposes (DPMM). To do so we will use the synthetic data generated based on Gaussian distributions. On this data, the clustering will be performed both by DPMM approach with the algorithm derived in the previous sections and the parameter fitting of typical finite mixture of gaussian (GMM). Although during the implementation, fitting to finite mixture of gaussian is applied with MATLAB's predefined functions, we will present the underlying algorithm in the appendix for the sake of completeness. For the experiments, MATLAB's **Statistics and Machine Learning Toolbox** is required.

5.1 Experiment Details

The experiments aims at demonstrating the clustering capability of the models based on Dirichlet Process. Our likelihood F is gaussian as mentioned above.

5.1.1 Data Generation

For the sake of visualization, we opted for working in 3-D data. It follows a generation of a typical generation from mixture of Gaussians: K denoting the number of clusters we obtain

$$g_k \sim \text{ Gamma(shape } = c, \text{ scale } = 1), \quad c = 5, \quad k = 1, \ldots, K$$

to be normalized thereafter to build the weights:

$$w_k = \frac{g_k}{\sum_{j=1}^{K} g_j}, \quad \sum_k w_k = 1$$

For the cluster means

$$\mu_k \in \mathbb{R}^d, \quad (\mu_k)_j \sim \text{Unif}(-4, 4) \text{ i.i.d. over } j = 1, \ldots, d.$$

and for the cluster covariances, anisotropic and rotated versions are preferred to avoid too trivial dataset: Random eigenvalues by scaling the base eigenvalues $\lambda^{\text{base}} = (3.0, 0.6, 0.3)$ element-wise

$$r_{k,j} \sim \text{Unif}(0.7, 1.3), \quad \lambda_{k,j} = \lambda_j^{\text{base}} \cdot r_{k,j}$$

Then we draw a random rotation basis $Q_k \in \mathbb{R}^{d \times d}$.

$$\Sigma_k = Q_k \, \text{diag}\left(\lambda_{k,1}, \ldots, \lambda_{k,d}\right) Q_k^\top + 0.10 I_d$$

which makes each cluster a rotated ellipsoid Gaussian with a small isotropic "jitter" term, ensuring also the numerical positive definitiveness. Hence, in the end, the final model to be sampled from is

$$x \sim \sum_{k=1}^{K} w_k \mathcal{N}\left(\mu_k, \Sigma_k\right)$$

5.1.2 Hyper-parameters of the Algorithm

The details are listed in the Table 5, presented in the Appendix.

5.1.3 Evaluation Metrics

The goal of evaluation is to assess the performance of the utilization of Dirichlet Process for the mixture model and of the finite mixture of Gaussian components (GMM).

BIC (Bayesian Information Criterion)

For a model M_K (a K-component GMM), BIC is defined as:

$$\mathrm{BIC}(K) = p_K \log N - 2 \log \hat{L}_K$$

- N : number of training points

- $\hat{L}_K = p\left(X \mid \hat{\theta}_K\right)$: maximized likelihood

- p_K : number of free parameters in the GMM

The number of free parameters of the model is calculated as follows:

$$p_K = (K-1) + Kd + K\frac{d(d+1)}{2}$$

- weights: $K-1$

- means: $K \cdot d$

- covariances: each covariance has $d(d+1)/2$ free parameters $\Rightarrow K \cdot d(d+1)/2$

Dahl Approach as Clustering for DPMM and as Model Selection for GMM

We have presented this approach in the previous sections for DPMM. It builds label-invariant approach to clustering. To promote the clustering consistency of the GMM and DPMM, we use Dahl's approach also for GMM as follows: For GMM, one needs to decide how many components to use as a hyperparameter. To do so we can take advantage of several model trials and fuse their information to get the final decision. For each fitted model m, you have a clustering $z^{(m)} \in \{1, \ldots, K_m\}^N$. Defining its association matrix

$$A_{ij}^{(m)} = \mathbf{1}\left\{z_i^{(m)} = z_j^{(m)}\right\}$$

Co-clustering matrix can be obtained by $S = \sum_m w_m A^{(m)}$ where the weights are based on BIC criterion:

$$w_m \propto \exp\left(-\frac{1}{2}\left(\mathrm{BIC}_m - \min_j \mathrm{BIC}_j\right)\right)$$

In the Dahl's approach, we pick the model based on

$$m^* = \arg\min_m \left\|A^{(m)} - S\right\|_F^2$$

which will make the clustering decision making (i.e. model selection for GMM).

During experiments we can also restrict the domain of the search to K_{mod}

$$m^*_{\mathrm{modeK}} = \arg \min_{m:K^{(m)}=K_{\mathrm{mode}}} \left\|A^{(m)} - S\right\|_F^2$$

where

$$W(K) = \sum_{m:K_m=K} w_m, \quad K_{\text{mode}} = \arg\max_K W(K).$$

Rand Index (RI) and Adapted Rand Index (ARI) [15]

Given a set of n elements $S = \{o_1, \ldots, o_n\}$ and two partitions of S to compare, $X = \{X_1, \ldots, X_r\}$, a partition of S into r subsets, and $Y = \{Y_1, \ldots, Y_s\}$, a partition of S into s subsets, define the following:

- a, the number of pairs of elements in S that are in the same subset in X and in the same subset in Y

- b, the number of pairs of elements in S that are in different subsets in X and in different subsets in Y

- c, the number of pairs of elements in S that are in the same subset in X and in different subsets in Y

- d, the number of pairs of elements in S that are in different subsets in X and in the same subset in Y

The Rand index, R, is:

$$R = \frac{a+b}{a+b+c+d} = \frac{a+b}{\binom{n}{2}}$$

This is equivalent to:

$$RI = \frac{TP+TN}{TP+FP+FN+TN}$$

which thus acts as a measure of the percentage of correct decisions made by the algorithm. ARI, on the other hand, is based on the contingency matrix defined as follows

X/Y	Y_1	Y_2	$\cdots$	Y_s	sums
X_1	n_{11}	n_{12}	$\cdots$	n_{1s}	a_1
X_2	n_{21}	n_{22}	$\cdots$	n_{2s}	a_2
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
X_r	n_{r1}	n_{r2}	$\cdots$	n_{rs}	a_r
sums	b_1	b_2	$\cdots$	b_s	

each entry n_{ij} denotes the number of objects in common between X_i and Y_j : $n_{ij} = |X_i \cap Y_j|$ and the index is defined as:

$$ARI = \frac{\sum_{ij}\binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2}\sum_j \binom{b_j}{2}\right]/\binom{n}{2}}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}\right] - \left[\sum_i \binom{a_i}{2}\sum_j \binom{b_j}{2}\right]/\binom{n}{2}}$$

where the interpretation is:

- ARI $= 1 \rightarrow$ perfect match

- ARI $\approx 0 \rightarrow$ about as good as random

- ARI $< 0 \rightarrow$ worse than random

In our case ARI and RI are constructed based on **true** labels and the **predicted** clustering labels on the same dataset

Per-Point Certainty

One of the aspects that DPMM enables us to do is to learn the certainty level of the clustering. Let $C(i)$ be the set of indices in the same Dahl cluster as point i. Point level certainty can be defined as:

$$c_i = \frac{1}{|C(i)| - 1} \sum_{j \in C(i), j \neq i} S_{ij}$$

- $c_i \in [0, 1]$

- $c_i \approx 1$: point i very consistently co-clusters with its assigned cluster-mates across posterior samples

- $c_i \approx 0$: point i is unstable/ambiguous: often not co-clustered with its assigned cluster-mates

Within-Cluster Variance

We have data points $x_1, \ldots, x_N \in \mathbb{R}^d$ and a clustering assignment $z_i \in \{1, \ldots, K\}$. We define the set for cluster k:

$$C_k = \{i : z_i = k\}, \quad n_k = |C_k|$$

For each cluster, to measure how "spread out" the points are around their mean we define $\text{SSE}_k = \sum_{i \in C_k} \|x_i - \mu_k\|^2$ and in total

$$\text{SSE} = \sum_{k=1}^{K} \text{SSE}_k = \sum_{k=1}^{K} \sum_{i \in C_k} \|x_i - \mu_k\|^2 .$$

Within cluster variance is then defined as $\text{WCV} = \frac{\text{SSE}}{N}$.

Total Predictive Log-likelihood on the Test Set

It aims at measuring the performance of the model on the test data by evaluating the likelihood on the test data, generated exactly from the same data generation model. Hence, the formula is $\sum_{i=1}^{N_{te}} \log p\left(x_i \mid X_{tr}\right)$

$$\begin{aligned} \text{predLL} &= \sum_{i=1}^{N_{te}} \log p\left(x_i \mid \text{ trained model }\right) \\ \text{avg predLL} &= \frac{1}{N_{te}} \text{ predLL}. \end{aligned}$$

for GMM model, it is simply:

$$\text{predLL}_{te}^{GMM} = \sum_{i=1}^{N_{te}} \log \left(\sum_{k=1}^{K} \hat{\pi}_k \mathcal{N}\left(x_i \mid \hat{\mu}_k, \hat{\Sigma}_k\right) \right)$$

For the DPMM, assume we have partitions for different iterations:

$$z^{(t)} = \left(z_1^{(t)}, \ldots, z_{N_{tr}}^{(t)}\right), \quad t = 1, \ldots, T$$

Our typical CRP formulation gives

$$p\left(x \mid X_{tr}, z^{(t)}, \alpha^{(t)}\right) = \sum_{k=1}^{K^{(t)}} \frac{n_k^{(t)}}{N_{tr} + \alpha^{(t)}} t_{\nu_k^{(t)}}\left(x \mid \mu_k^{(t)}, \Sigma_k^{(t)}\right) + \frac{\alpha^{(t)}}{N_{tr} + \alpha^{(t)}} t_{\nu_0}\left(x \mid \mu_0, \Sigma_0\right)$$

where t_ν denotes the Student-t distribution. Since after thinning and burn-in, we claim that these samples are representative of the intended posterior, we can use the Monte-Carlo predictive as

$$p\left(x \mid X_{tr}\right) \approx \frac{1}{T} \sum_{t=1}^{T} p\left(x \mid X_{tr}, z^{(t)}, \alpha^{(t)}\right)$$

Therefore.

$$\text{predLL}_{te}^{DP} = \sum_{i=1}^{N_{te}} \log p\left(x_i \mid X_{tr}\right) \approx \sum_{i=1}^{N_{te}} \log\left(\frac{1}{T} \sum_{t=1}^{T} p\left(x_i \mid X_{tr}, z^{(t)}, \alpha^{(t)}\right)\right)$$

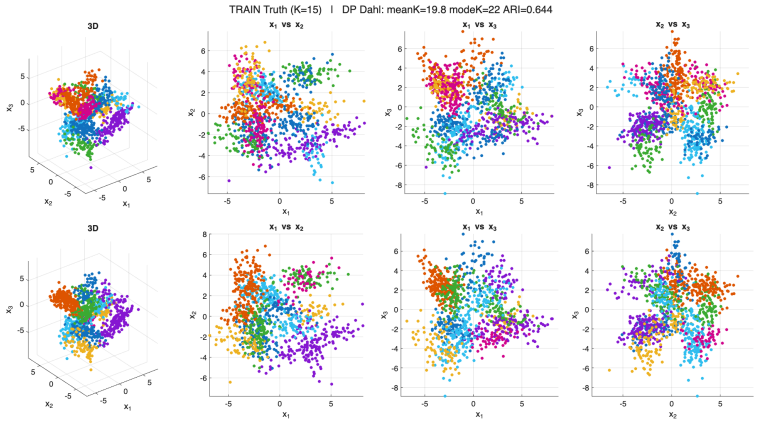
5.2 Experiment Results

5.2.1 K=15

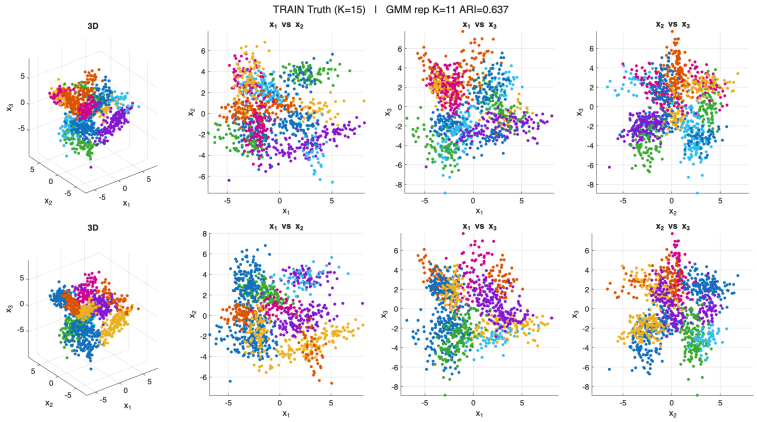
Now we analyze 15 different cluster case. The generated dataset conforms with the values of the Table 5. The visual comparison between the true versus predicted clusters for DPMM and GMM are shown in the Fig 1 and the Table 3. Based on these results, we can see that DPMM performs even slightly better than the GMM models' decision fusion (which is based on GMM models where the number of tried mixture components ranges from -5 to +5 around the true model's number of cluster in order to emulate the scenario where we don't know the true number of clusters but we somehow can find an interval sandwiching the true value. The fusion is based on BIC weighted Dahl's approach as discussed above). The performance of DPMM is slightly higher for both training data-points and test likelihood unsupervised performance. Furthermore, the evolution of DP clustering along with the α parameter can be observed via different snapshots, some of which are depicted in the Fig 2. The stability of the posterior $\log p(z, X \mid \alpha$, NIW hyperparams) can be seen by the Fig 3

Table 3: GMM vs. DP-MM results K=15 (train/test).

Metric	GMM sweep + Dahl–BIC	DP-MM (Dahl ; full Bayes test)
K (mean / mode)	11	19.770 / 22
α (mean / median)	–	3.122 / 3.059
RI (train)	0.940	0.948
ARI (train)	0.637	0.644
WCV (train)	4.775	4.258
Test predLL (avg)	−7448.93 (−6.6213)	−7382.96 (−6.5626)

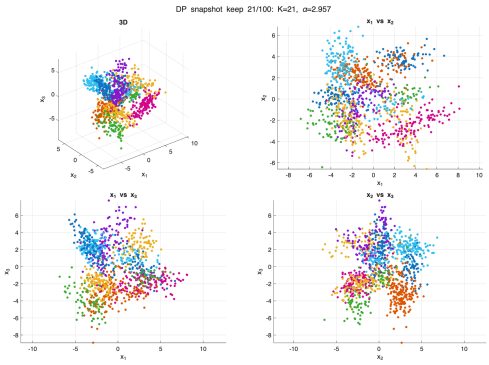


(a) True (Top) vs DPMM (Bottom) Cluster Prediction

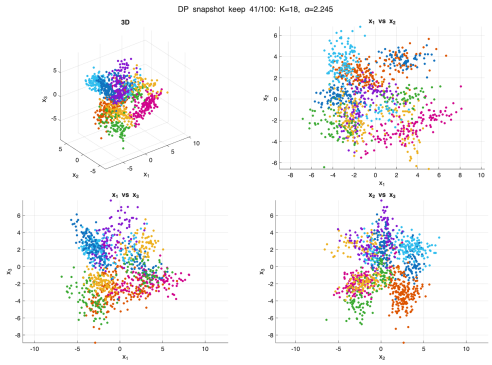


(b) True (Top) vs GMM (Bottom) Cluster Prediction

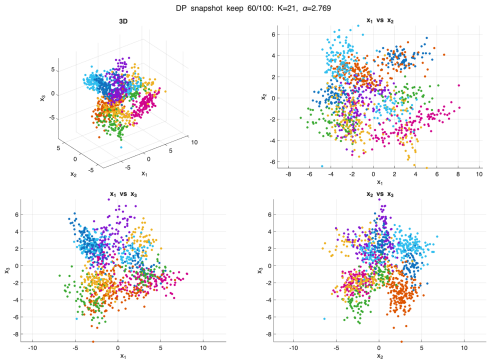
Figure 1: Visual Comparison Between GMM and DPMM.



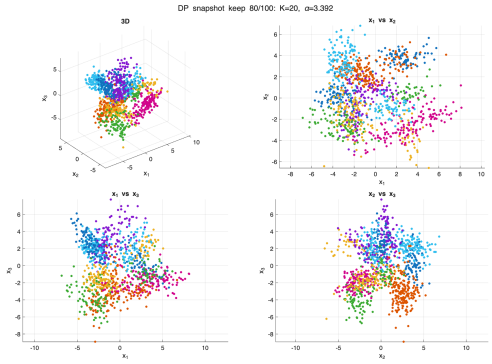
(a) Snapshot 21: $\alpha = 2.957$



(b) Snapshot 41: $\alpha = 2.245$



(c) Snapshot 60: $\alpha = 2.769$



(d) Snapshot 80: $\alpha = 3.392$

Figure 2: Evolution of DP Snapshots. Top row: Iterations 21 and 41. Bottom row: Iterations 60 and 80.

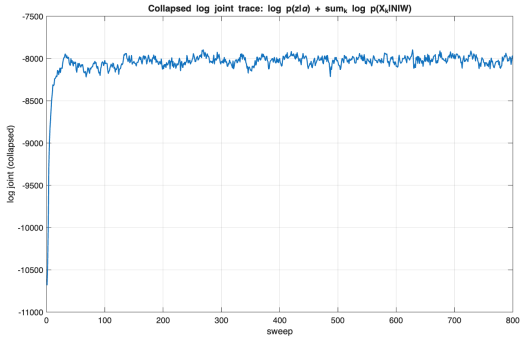


Figure 3: Joint Distribution Value in Log Domain

Another advantage of DP is that it provides us with the uncertainty of the clustering since the distribution is obtained. Practically the samples $z^{(1)}, z^{(2)}, \ldots, z^{(T)}$ provide the indicator matrix $A_{ij}^{(t)} = \mathbf{1}\left[z_i^{(t)} = z_j^{(t)}\right]$, whereby the co-clustering matrix (sorted) is built

$$S_{ij} = \mathbb{P}\left(z_i = z_j \mid X\right) \approx \frac{1}{T}\sum_{t=1}^{T} A_{ij}^{(t)} = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\left[z_i^{(t)} = z_j^{(t)}\right]$$

The true clustering map and the heatmap comparison can be seen in Fig 4 which indicates the diagonal pattern is majorly obtained. However, the parceling of main squares can indicate intermittent split-merge of the clusters through iterations. Moreover, thin vertical or horizontal lines at off-diagonal regions denote uncertain clustering of certain points. Additionally, the clustering certainty can be shown by the certainty

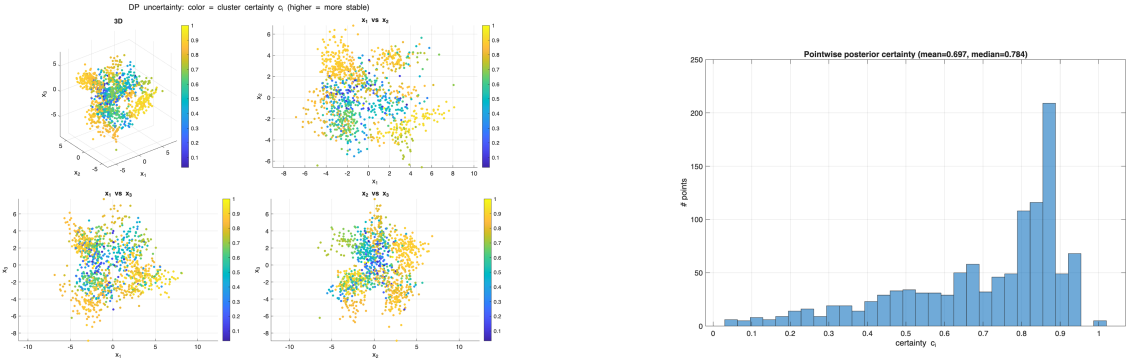
scatter and histogram plots as seen in Fig 5, whereby we observe that the DP clustering is around high precision of many points.



(a) DPMM Prediction Co-clustering HeatMap

(b) True Co-Clustering

Figure 4: Co-Clustering Performance



(a) Uncertainty Map on Scatter Graph

(b) Histogram of the Uncertainty Value

Figure 5: Uncertainty in Clustering

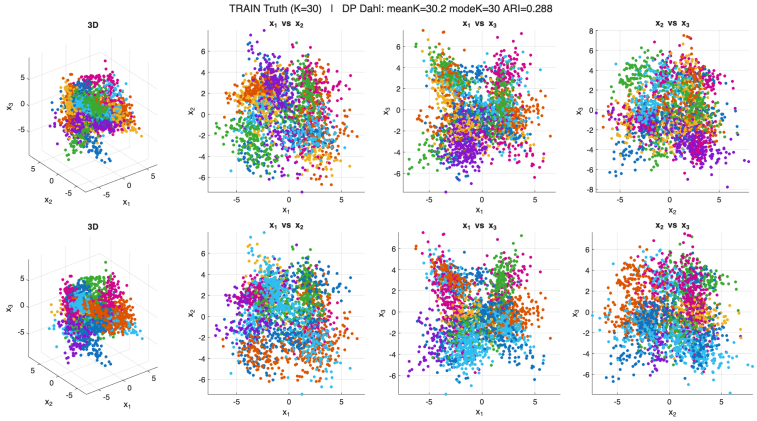
5.2.2 K=30

Now we double the number of clusters of the generated dataset. In that scenario, data becomes much more interleaved. The tabular results are depicted in the Table 4. Low ARI and high RI for both GMM and DPMM signifies that they correctly captured most of the different pairs but same cluster pairs are wrong, which is likely to stem from complex generation of the data with hard separation. These clustering metrics are slightly lower for DPMM, whereas the unsupervised performance of the DPMM (based on the likelihood) on the test data is higher, which shows its generalization. Moreover, its estimated number of clusters are so close to the real one, demonstrating its capability to capture the underlying clusters. Visual comparison between GMM and DPMM is shown in Fig 6. The convergence of the posterior is shown in Fig 8. Within cluster variance (WCV) is larger for DPMM, which can be explained by the fact that true clusters too are of elongated shapes. Thus, DP captures better the original shapes and does not try to fit to concise blobs as GMM.

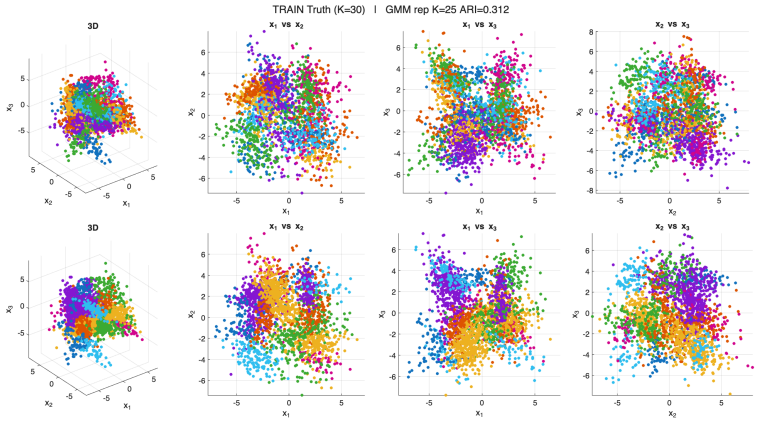
As can be seen in the Fig 7, during the evolution of the samples, the concentration parameter α values remain higher compared to $K = 15$ case as expected, which indicates that our updates in Escobar approach within MCMC samples, enables the model to adapt itself to accommodate higher number of clusters. Therefore, it shows the versatility of DPMM model, whereas for classical GMM, we are bound with initial hypothesis to select the range of multiple models to be tried.

Table 4: GMM vs. DP-MM results K=30 (train/test).

Metric	GMM sweep + Dahl-BIC	DP-MM (Dahl ; full Bayes test)
K (mean / mode)	25	30.210/30
α (mean / median)	–	4.363/4.259
RI (train)	0.935	0.931
ARI (train)	0.312	0.288
WCV	4.270	4.856
Test predLL (avg)	−6.7629	−6.6918



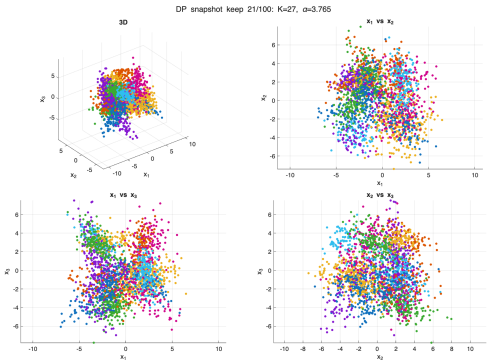
(a) True (Top) vs DPMM (Bottom) Cluster Prediction



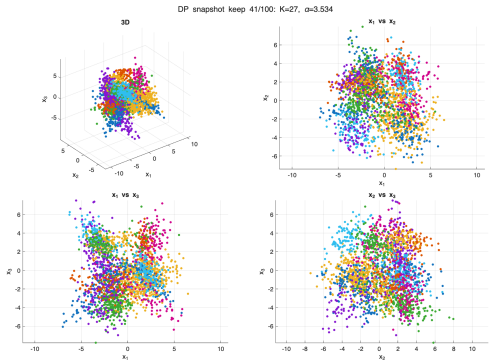
(b) True (Top) vs GMM (Bottom) Cluster Prediction

Figure 6: Visual Comparison Between GMM and DPMM for $K = 30$.

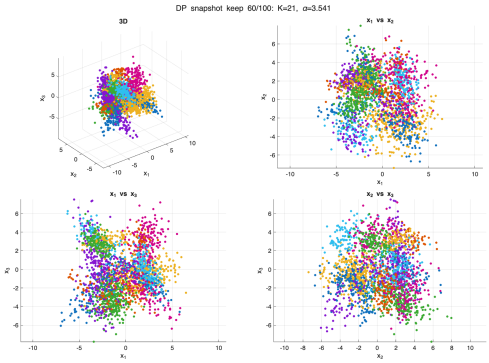
Furthermore, when compared with the previous scenario, the uncertainty of clustering, (captured by the co-clustering matrix and the histogram in Fig 9) is much higher for $K = 30$ which corroborates to the low ARI result and can be explained by increased complexity of the incoming data to be clustered



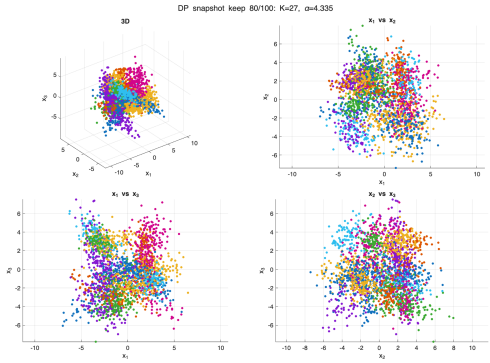
(a) Snapshot 21: $\alpha = 3.765$



(b) Snapshot 41: $\alpha = 3.534$



(c) Snapshot 60: $\alpha = 3.541$



(d) Snapshot 80: $\alpha = 4.335$

Figure 7: Evolution of DP Snapshots. Top row: Iterations 21 and 41. Bottom row: Iterations 60 and 80.

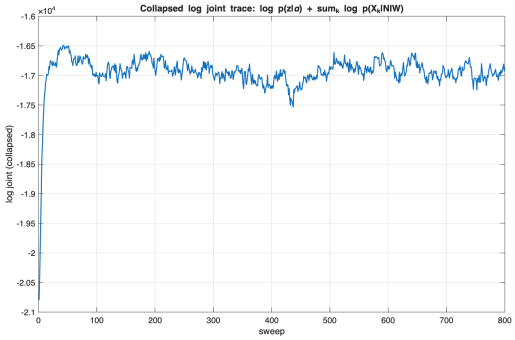
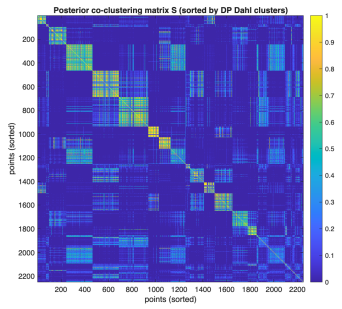


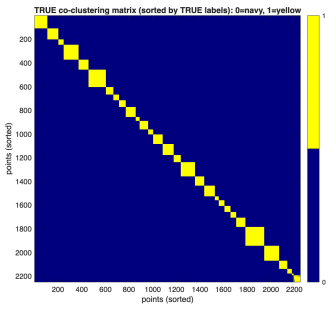
Figure 8: Joint Distribution Value in Log Domain

5.2.3 Other Experiments

The other experiments with different data generation model uses again $K = 15$ and $K = 30$ and their detailed results are presented in the Appendix. However, we can conclude that despite different data generation

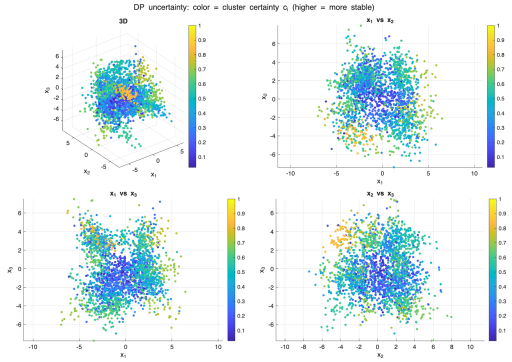


(a) DPMM Prediction Co-clustering HeatMap

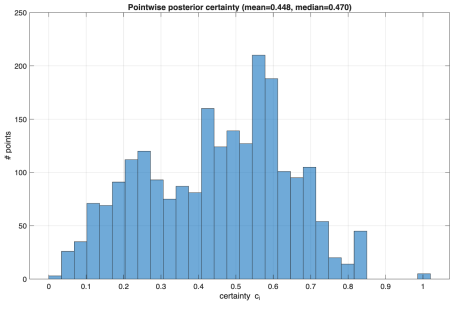


(b) True Co-Clustering

Figure 9: Co-Clustering Performance $K = 30$



(a) Uncertainty Map on Scatter Graph



(b) Histogram of the Uncertainty Value

Figure 10: Uncertainty in Clustering $K = 30$

6 Conclusion

In this project, we developed a measure-theoretic and constructive description of the Dirichlet Process (DP) as a distribution over probability measures, and then utilized that viewpoint to motivate Dirichlet Process Mixture Models (DPMMs) as a principled Bayesian nonparametric alternative approach to classical finite mixtures, whereby connecting Ferguson's definition, Sethuraman's stick-breaking construction, and the Chinese Restaurant Process into a coherent clustering. While Gaussian Mixture Models (GMMs) necessitates fixing the number of components and often rely on repeated model fitting for selection, the DP prior allows model complexity to be inferred through Bayesian updating.

Based on our experiments, empirically, on synthetic data with $K = 15$, DPMM is observed to achieve stronger agreement with ground truth than the BIC-weighted GMM sweep (e.g., higher RI/ARI), while also providing slightly better predictive test log-likelihood, demonstrating both improved clustering fidelity on so-called training data and competitive density generalization (test data). For the more challenging, imbalanced $K = 30$ case scenario, the DPMM's posterior favored a smaller effective number of occupied clusters (reflecting the merging of tiny/empty components), which can reduce ARI under label-based evaluation, yet still improved predictive likelihood on held-out data highlighting the practical trade-off between capturing each and every cluster versus capturing the dominant data-generating structure.

Beyond these properties, DPMM naturally captures the uncertainty and its quantification of the clustering through various samples obtained from the posterior label distribution. Hence, it offers a diagnostic insight into ambiguous regions where split/merge behavior occurs across iterations that are fundamentally absent from single-fit parametric point-estimate solutions such as EM-GMM.

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7 Appendix

The high quality original report file and the codes and results can be found in the following link: https://drive.google.com/drive/folders/1GcmzFu0bfQabeduRTHmy_CeZTFLnP76S?usp=sharing

7.1 Algorithm Implementation Related Details

Table 5: Hyperparameters and fixed settings for the DP vs GMM same-generator likelihood + uncertainty experiment.

Block	**Parameter**	**Value**	**Notes**
Global	Dimension d	3	3D synthetic data
Global	True clusters K_{true}	30	Used by generator and fixed-K GMM
Global	Train size N_{tr}	$K \times 75$	Equal per-cluster nominal scaling
Global	Test size N_{te}	$K \times 75$	Same generator as train
Generator	Weight concentration	`dir_conc` $= 5$	Draw $w_k \sim \text{Gamma}(5, 1)$ then normalize
Generator	Means range	$\approx [-4, 4]^3$	$\mu = 4(2u - 1)$ with $u \sim \text{Unif}(0, 1)$
Generator	Base eigvals	[3.0, 0.6, 0.3]	Anisotropic covariances
Generator	Eig scaling factor	$0.7 + 0.6u$	Per-dim factor in $[0.7, 1.3]$
Generator	Cov jitter	$0.10I$	Ensures SPD / moderate overlap
GMM (all)	Covariance type	full	`'CovarianceType','full'`
GMM (all)	Regularization	10^{-6}	`'RegularizationValue',1e-6`
GMM (all)	Replicates	5	Multiple EM restarts
GMM (all)	Max EM iters	1000	`statset('MaxIter',1000)`
GMM fixed-K	K	30	$K = K_{\text{true}}$
GMM sweep	K range	25–35	$K_{\min} = K_{\text{true}} - 5$, $K_{\max} = K_{\text{true}} + 5$
DP/NIW prior	κ_0	1.0	Mean strength
DP/NIW prior	ν_0	$d + 5 = 8$	Degrees of freedom
DP/NIW prior	Λ_0	$5I$	Scale matrix
DP α prior	a_α (shape)	2.0	$\alpha \sim \text{Gamma}(a_\alpha, \text{rate} = b_\alpha)$
DP α prior	b_α (rate)	1.0	Rate parameter
DP init	α_{init}	1.0	Initial concentration
DP init	K_{init}	8	$\min(8, \max(2, \text{round}(\sqrt{N/10})))$
DP sampler	Total sweeps	800	`T_total`
DP sampler	Burn-in	400	`burn_in`
DP sampler	Thinning	4	`thin`
DP sampler	Kept samples	100	$(800 - 400)/4$
DP test LL	Freeze-α summary	mean	`alpha_summary='mean'`
DP test LL	$\hat{\alpha}$	$\text{mean}(\alpha_{\text{keep}})$	Used in freeze-α predictive
DP test LL	Full Bayes	$\alpha_t = \alpha_{\text{keep}}(t)$	Average predictive over (Z_t, α_t)

7.2 Additional Experiment No1

7.2.1 Dataset Generation

Mean

To perform this experiment, the data generation is changed as follows;

The gamma distribution is used again for the prior on the components' weights. Moreover, we build the components around 2 circularly placed mean vectors by splitting K components into two sets $K_1 = \lceil K/2 \rceil, K_2 = K - K_1$

$$\mu_j^{(1)} = \begin{bmatrix} R_1 \cos t_j^{(1)} \\ R_1 \sin t_j^{(1)} \\ z_j^{(1)} \end{bmatrix}$$

where

- angles $t_j^{(1)}$ evenly spaced on $[0, 2\pi)$

- radius $R_1 = 4.0$

- z -coordinates linearly spread $[-1.8, 1.8]$

and

$$\mu_j^{(2)} = \begin{bmatrix} R_2 \cos t_j^{(2)} \\ R_2 \sin t_j^{(2)} \\ z_j^{(2)} \end{bmatrix}$$

where

- angles shifted by $\pi/6$

- radius $R_2 = 3.3$

- z-coordinates linearly spread $[1.2, -1.2]$

and the gaussian jitter is added

Covariance

3 major families of covariance shapes can be defined:

- Needle: [5.0, 0.18, 0.10]

- Sheet: [2.0, 1.3, 0.12]

- Round: [1.4, 0.9, 0.6]

and for each component, we choose one of them (calling it a base vector $e^{(0)} \in \mathbb{R}^3$.)

We perturb these base vectors

$$e_{k,i} = s_k e_i^{(0)} \exp\left(0.35\xi_{k,i}\right), \quad \xi_{k,i} \sim \mathcal{N}(0,1)$$

based on $s_k \sim \text{Uniform}(\ 0.35, 3.15)$

Similar to our original dataset, we build the orthogonal matrix by QR decomposition of a random matrix

$$A_k \in \mathbb{R}^{3\times 3}, (A_k)_{ij} \sim \mathcal{N}(0,1), \quad A_k = Q_k R_k$$

$$\Sigma_k = Q_k \,\text{diag}\left(e_{k,1}, e_{k,2}, e_{k,3}\right) Q_k^\top + 0.05 I_3$$

The details are tabulated in Table 6.

Table 6: Synthetic interleaved Gaussian-mixture generator (3D).

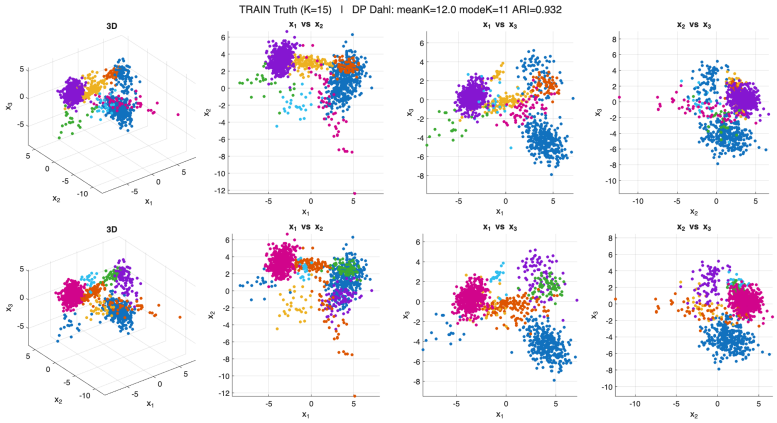
Component	**Parameterization in the code**
Dimension	$d = 3$
Mixture weights	
Gamma draw (unnormalized)	$\tilde{w}_k \sim \text{Gamma}(\text{shape} = 0.6,\ \text{scale} = 1)$
Normalization	$w_k = \tilde{w}_k / \sum_{\ell=1}^{K} \tilde{w}_\ell$
Means (two rings + jitter)	
Split	$K_1 = \lceil K/2 \rceil, \quad K_2 = K - K_1$
Ring 1 radius	$R_1 = 4.0$
Ring 2 radius	$R_2 = 3.3$
Angles	$t^{(1)} : \text{linspace}(0, 2\pi, K_1), \quad t^{(2)} : \text{linspace}(\pi/6, 2\pi + \pi/6, K_2)$
z-spread (ring 1)	$z^{(1)} : \text{linspace}(-1.8,\ 1.8,\ K_1)$
z-spread (ring 2)	$z^{(2)} : \text{linspace}(1.2,\ -1.2,\ K_2)$
Mean jitter	$\mu_k \leftarrow \mu_k + \epsilon_k, \quad \epsilon_k \sim \mathcal{N}(0,\ 1.3^2 I_3)$
Covariances (anisotropic + random rotation)	
Shape family probabilities	Needle / Sheet / Round chosen by $u \sim \text{Unif}(0,1)$ with thresholds $0.34, 0.67$
Base eigenvalues (needle)	$[5.0,\ 0.18,\ 0.10]$
Base eigenvalues (sheet)	$[2.0,\ 1.3,\ 0.12]$
Base eigenvalues (round-ish)	$[1.4,\ 0.9,\ 0.6]$
Global scale	$s_k = 0.35 + 2.8\,u, \quad u \sim \text{Unif}(0,1)$
Eigen perturbation	$e_{k,i} = s_k e_i^{(0)} \exp(0.35\,\xi_{k,i}),\ \xi_{k,i} \sim \mathcal{N}(0,1)$
Random rotation	$A_k \sim \mathcal{N}(0,1)^{3\times 3}, \quad [Q_k, \sim] = \text{qr}(A_k)$
Covariance construction	$\Sigma_k = Q_k \text{diag}(e_{k,1}, e_{k,2}, e_{k,3}) Q_k^\top + 0.05 I_3$
Sampling	
Cluster counts	$(n_1, \ldots, n_K) \sim \text{Multinomial}(N; w_1, \ldots, w_K)$
Data draw	$x \mid (z = k) \sim \mathcal{N}(\mu_k, \Sigma_k)$

7.2.2 Experiment Results for $K = 15$

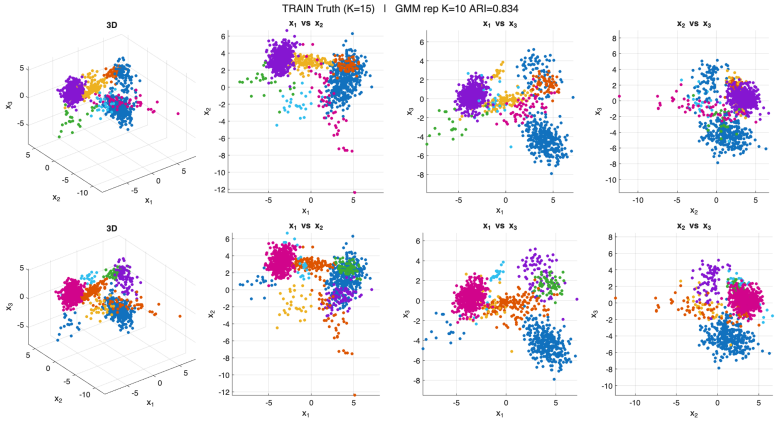
As can be seen in Table 7, Fig 11, Fig 12, Fig 13, Fig 14 and Fig 15, DPMM can cluster the data with higher performance metrics than classical GMM parameter fit with high precision.

Table 7: Experiment summary (GMM vs DP).

	GMM sweep + Dahl-BIC rep	**DP (alpha updated + fullBayes test)**
Sweep K range	$[10, 20]$	—
Representative K	10	$\mathbb{E}[K] = 12.020,\ \text{mode}(K) = 11$
$\mathbb{E}[\alpha]$ / median(α)	—	1.827 / 1.796
RI	0.942	0.975
ARI	0.834	0.932
WCV	3.358	3.528
TEST predLL (total)	−6448.08	−6425.51
TEST predLL (avg)	−5.7316	−5.7116



(a) True (Top) vs DPMM (Bottom) Cluster Prediction

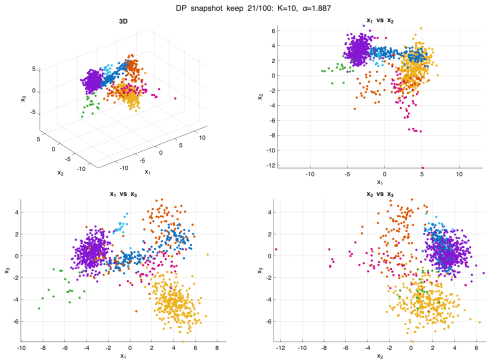


(b) True (Top) vs GMM (Bottom) Cluster Prediction

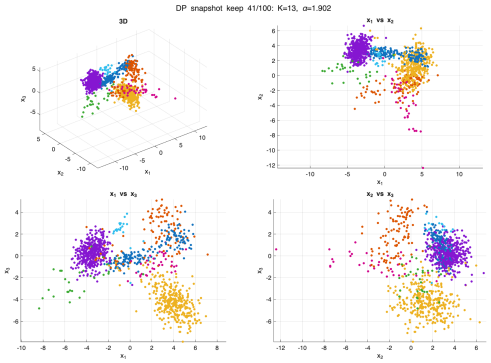
Figure 11: Visual Comparison Between GMM and DPMM for $K = 15$.

7.2.3 Experiment Results for $K = 30$

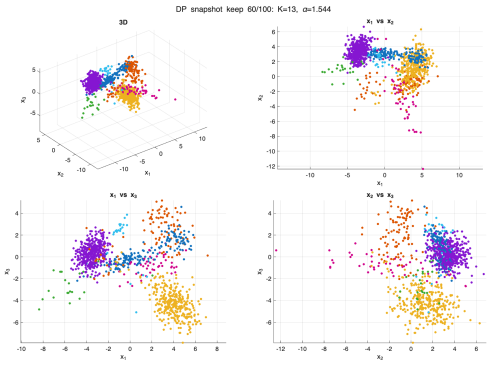
The Table 8, Fig 16, 17, 19, 20 and 18 summarizes the results. Here, mainly due to large imbalance in the number of representative points of the clusters, the number of clusters discovered by the DPMM remains at the level of 17-18. (In fact, when checked, only 26/30 of the clusters are non empty in the training and 28/30 are non-empty in the test case. Hence, the model supports at most 30 clusters but not all of them are represented. Moreover, 8 clusters are represented less than 5 points. Hence, it can be concluded that some of the tiny clusters are merged under the DP model). Such an imbalance results in deterioration in the performance in ARI. Nonetheless, for the likelihood generalization, DPMM performs still slightly better than the typical GMM



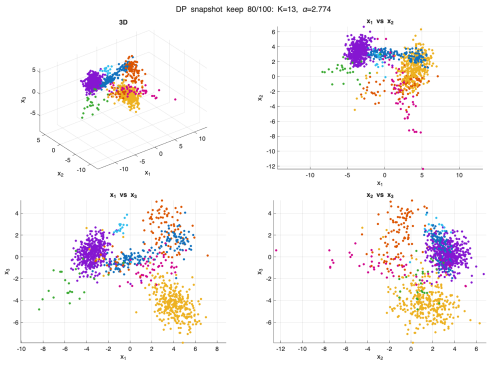
(a) Snapshot 21: $\alpha = 1.887$



(b) Snapshot 41: $\alpha = 1.902$



(c) Snapshot 60: $\alpha = 1.544$



(d) Snapshot 80: $\alpha = 2.774$

Figure 12: Evolution of DP Snapshots. Top row: Iterations 21 and 41. Bottom row: Iterations 60 and 80.

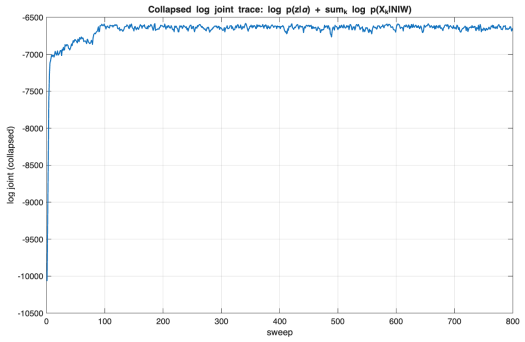


Figure 13: Joint Distribution Value in Log Domain $K = 15$

Table 8: Experiment summary (GMM Dahl–BIC representative vs. DP–GMM).

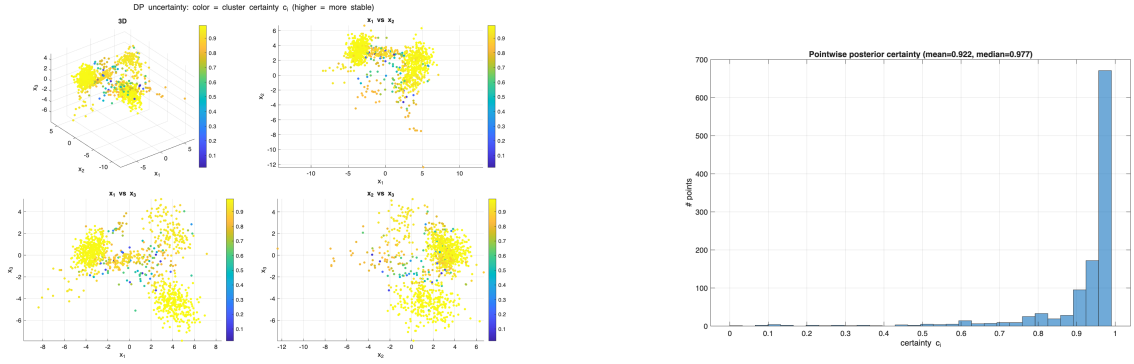
Metric	GMM (Dahl–BIC rep)	DP (Dahl rep)
Selected / implied K	25	17.61 (mode 17)
Train RI	0.929	0.913
Train ARI	0.537	0.589
Train WCV	8.237	9.139
Test predLL (avg)	-15042.64 (-6.6856)	-14842.97 (-6.5969)
Mean α (DP)	–	2.467
Median α (DP)	–	2.313



(a) DPMM Prediction Co-clustering HeatMap

(b) True Co-Clustering

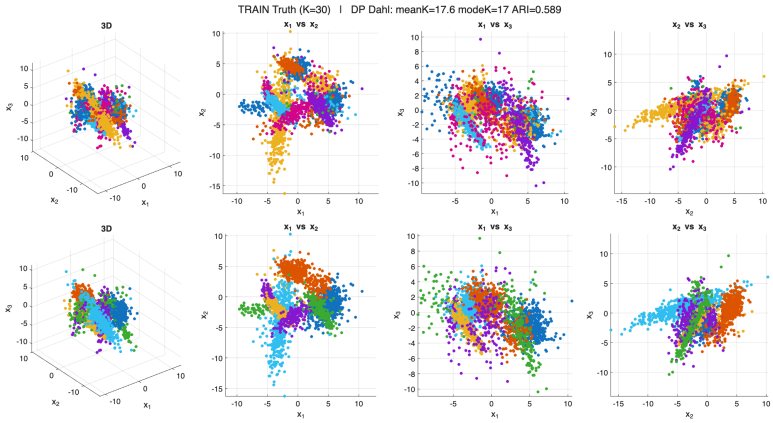
Figure 14: Co-Clustering Performance $K = 15$



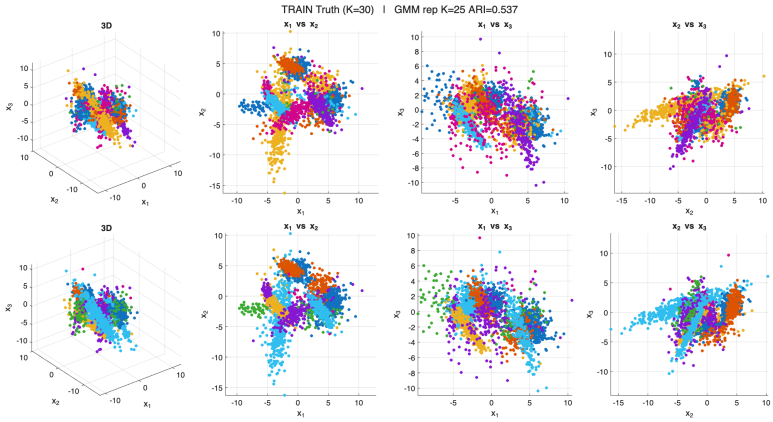
(a) Uncertainty Map on Scatter Graph

(b) Histogram of the Uncertainty Value

Figure 15: Uncertainty in Clustering $K = 15$

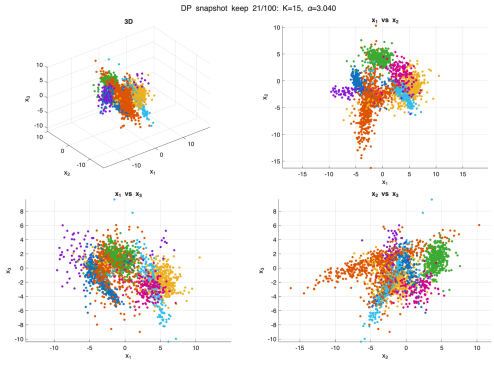


(a) True (Top) vs DPMM (Bottom) Cluster Prediction

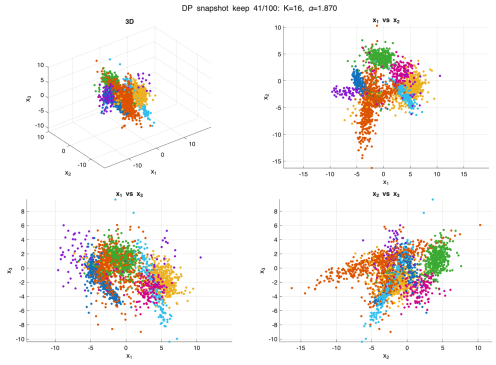


(b) True (Top) vs GMM (Bottom) Cluster Prediction

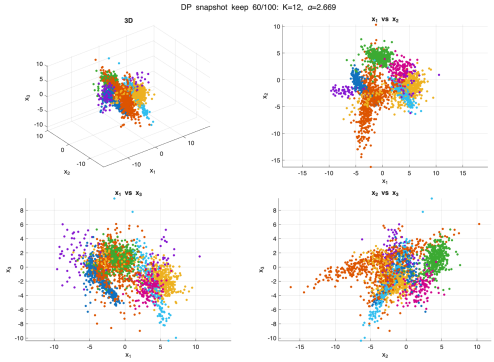
Figure 16: Visual Comparison Between GMM and DPMM for $K = 30$.



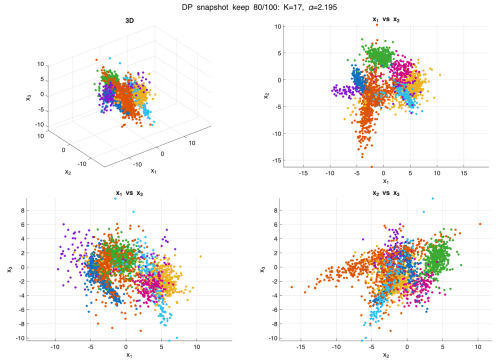
(a) Snapshot 21: $\alpha = 3.040$



(b) Snapshot 41: $\alpha = 1.870$



(c) Snapshot 60: $\alpha = 2.669$



(d) Snapshot 80: $\alpha = 2.195$

Figure 17: Evolution of DP Snapshots. Top row: Iterations 21 and 41. Bottom row: Iterations 60 and 80.

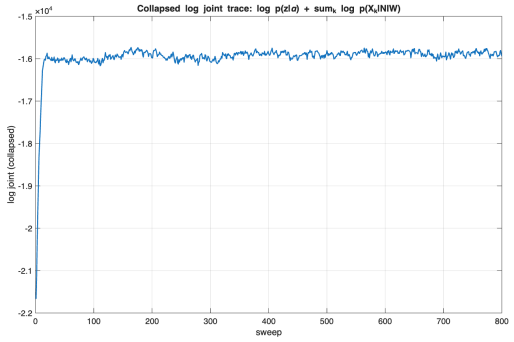
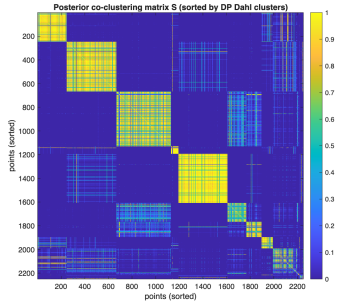
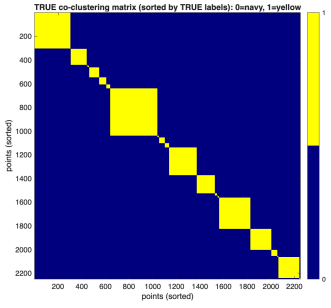


Figure 18: Joint Distribution Value in Log Domain $K = 30$

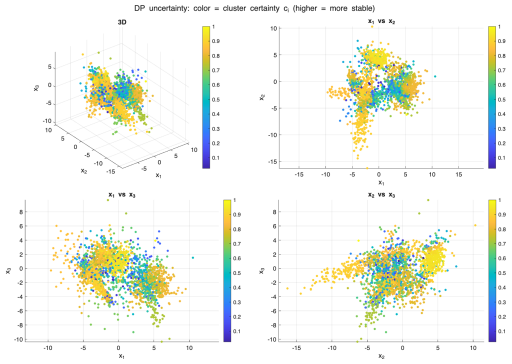


(a) DPMM Prediction Co-clustering HeatMap

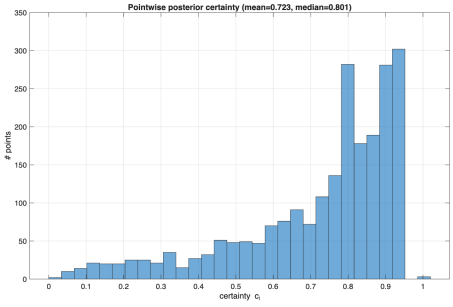


(b) True Co-Clustering

Figure 19: Co-Clustering Performance $K = 30$



(a) Uncertainty Map on Scatter Graph



(b) Histogram of the Uncertainty Value

Figure 20: Uncertainty in Clustering $K = 30$

7.3 GMM Fitting Algorithm Via Expectation Maximization

For the mixture model setup, we define the latent (unknown) cluster variable $z \in \{1, \ldots, k\}$ which defines

$$p(x, z \mid \theta) = p(z \mid \theta)p(x \mid z, \theta)$$

For gaussian mixture model, the parameter set is

$$\theta = \{\pi_c, \mu_c, \Sigma_c\}_{c=1}^{k}, \quad \pi_c \geq 0, \sum_c \pi_c = 1, \Sigma_c \succ 0$$

$$p(z = c \mid \theta) = \pi_c, \quad p(x \mid z = c, \theta) = \mathcal{N}(x \mid \mu_c, \Sigma_c)$$

Hence,

$$p(x \mid \theta) = \sum_{c=1}^{k} \pi_c \mathcal{N}(x \mid \mu_c, \Sigma_c)$$

the ultimate goal is

$$\max_{\theta} \log p(X \mid \theta) = \sum_{i=1}^{n} \log \left(\sum_{c=1}^{k} \pi_c \mathcal{N}(x_i \mid \mu_c, \Sigma_c) \right)$$

which is not easy to solve in that form. Therefore, variational form is used:

Defining

$$\Delta^{k-1} \triangleq \left\{ \vec{q} = [q(1), \cdots q(k)] \text{ s.t. } q(c) \geqslant 0 \text{ and } \sum_{c=1}^{k} q(c) = 1 \right\}$$

The claim is that $-\ln(p(x \mid \theta)) = \min_{q \in \Delta^{k-1}} \sum_{c=1}^{k} q(c)[-\ln(p(x, z = c \mid \theta) + \ln(q(c))]$ and the minimizer $q(c)$ is $q(c) = p(z = c \mid x, \theta)$

7.3.1 Proof

$$\nabla_{q(c)} \left(\sum_{-1}^{k} q(c) \left[-\ln(p(x, z = c \mid \theta) + \ln(q(c)) \right] + \lambda \left(\sum_{c=1}^{k} q(c) - 1 \right) \right) = 0.$$

$$\begin{aligned} &-\ln\left(p(x, z = c \mid \theta) + 1 \cdot \ln(q(c)) + \frac{1}{q(c)} q(c) + \lambda = 0. \right. \\ &\ln(q(c)) = \ln(p(x, z = c \mid \theta)) + (-1 - \lambda) \\ &q(c) = p(x, z = c \mid \theta) e^{-(\lambda+1)} \end{aligned}$$

Knowing that

$$\sum_{c=1}^{h} q(c) = e^{-(\lambda+1)} \sum_{c=1}^{k} p(x, z = c \mid \theta) = e^{-(\lambda+1)} p(x \mid \theta)$$
$$= 1$$

$p(x \mid \theta) = e^{-t(\lambda+1)}$ which then gives

$$q(c) = \frac{p(x, z = c \mid \theta)}{p(x \mid \theta)} = p(z = c \mid x, \theta)$$

putting this $q(c)$ into $\sum_{c=1}^{k} q(c)[-\ln(p(x, z = c \mid \theta) + \ln(q(c))]$

$$\sum_{c=1}^{k} p(z = c \mid x, \theta)[-\ln(p(x, z = c \mid \theta)) + \ln(p(z = c \mid \pi, \theta))]$$
$$= -\sum_{c=1}^{k} p(z = c \mid x, \theta) \ln\left(\frac{p(x, z = c \mid \theta)}{p(x, z = c \mid \theta)} p(x \mid \theta)\right)$$

$$= -\ln(p(x \mid \theta)) \sum_{c=1}^{z} p(z = c \mid x, \theta) = -\ln(p(x \mid \theta))$$

due to normalization. Therefore, minimizing $\sum_{i=1}^{r} -\ln\left(p\left(x^{(i)}(\theta)\right)\right.$ is the same as

$$\sum_{i=1}^{n} \sum_{c=1}^{k} p\left(z = c \mid x^{(i)}, \theta\right) \left[-\ln\left(p\left(x^{(i)}, z = c \mid \theta\right) + \ln\left(p\left(z = c \mid x^{(i)}, \theta\right)\right)\right)\right]$$

$$= \sum_{i=1}^{n} \sum_{c=1}^{k} p\left(z = c \mid x^{(i)}, \theta\right) \left[-\ln\left(p(z = c \mid \theta) - \ln\left(p\left(x^{(i)} \mid z, \theta\right)\right)\right)\right.$$
$$- \sum_{j=1}^{n} \sum_{c=1}^{k} -p\left(z = c \mid x^{(i)}, \theta\right) \ln\left(p\left(z = c \mid x^{(i)}, \theta\right)\right)$$

$$= \sum_{i=1}^{n} \sum_{c=1}^{k} p\left(z = c \mid x^{(i)}, \theta\right) \left[-\ln \mathrm{p}(z = c \mid \theta) - \ln\left(p\left(x^{(i)} \mid z, \theta\right)\right)\right.$$
$$- \sum_{i=1}^{n} - \sum_{c=1}^{k} q_i(c) \ln(q_i(c))$$

However, now $q(c)$ is already optimized and the optimization with respect to θ is irrelevant of q. Hence:

$$\min_{\theta} \sum_{i=1}^{n} -\ln p\left(x^{(i)} \mid \theta\right) \equiv \min_{\theta} \sum_{i=1}^{n} \underbrace{\min_{q_i \in \Delta^{k-1}} \sum_{c=1}^{k} q_i(c) \left[-\ln p\left(x^{(i)}, z = c \mid \theta\right) + \ln q_i(c)\right]}_{\text{variational lemma for each } x^{(i)}}$$

Hence, the expectation-maximization algorithm is as follows:

(E-Step): Fix θ, solve $\{q_i\}_{i=1}^{n}$ as

$$q_i(c) = p\left(z = c \mid \theta, x^{(i)}\right).$$

(M-Step): Fix $\{q_i\}_{i=1}^n$, solve weighted joint MLE

$$\min_\theta \sum_{i=1}^n \sum_{c=1}^k q_i(c) \left[-\ln q\left(x^{(i)}, z = c \mid \theta\right)\right]$$

GMM

When calculations are done for $q_i(c) \leftarrow p\left(z_i = c \mid x_i, \theta\right)$ and then $\theta \leftarrow \arg\max_\theta \sum_{i=1}^n \sum_{c=1}^k q_i(c) \ln p\left(x_i, z_i = c \mid \theta\right)$, [16], we obtain the following EM steps:

E-step

$$r_{ic} \leftarrow \frac{\pi_c \mathcal{N}\left(x_i \mid \mu_c, \Sigma_c\right)}{\sum_{j=1}^k \pi_j \mathcal{N}\left(x_i \mid \mu_j, \Sigma_j\right)}$$

M-step

$$N_c \leftarrow \sum_i r_{ic}, \quad \pi_c \leftarrow \frac{N_c}{n}, \quad \mu_c \leftarrow \frac{1}{N_c} \sum_i r_{ic} x_i$$
$$\Sigma_c \leftarrow \frac{1}{N_c} \sum_i r_{ic} \left(x_i - \mu_c\right) \left(x_i - \mu_c\right)^\top$$